Guana Tolomato Matanzas National Estuarine Research Reserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

Contents

Funding & Acknowledgements

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m^-1	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

SEACAR QAQC Description	*Include*	*SEACAR QAQCFlagCode*
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

Qualifier Source	*Value Qualifier*	*Include*	*MDL*	*Description*
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determiniation; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

Qualifier Source	*Value Qualifier*	*Include*	*Description*
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as El Niño/La Niña, large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p >= 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

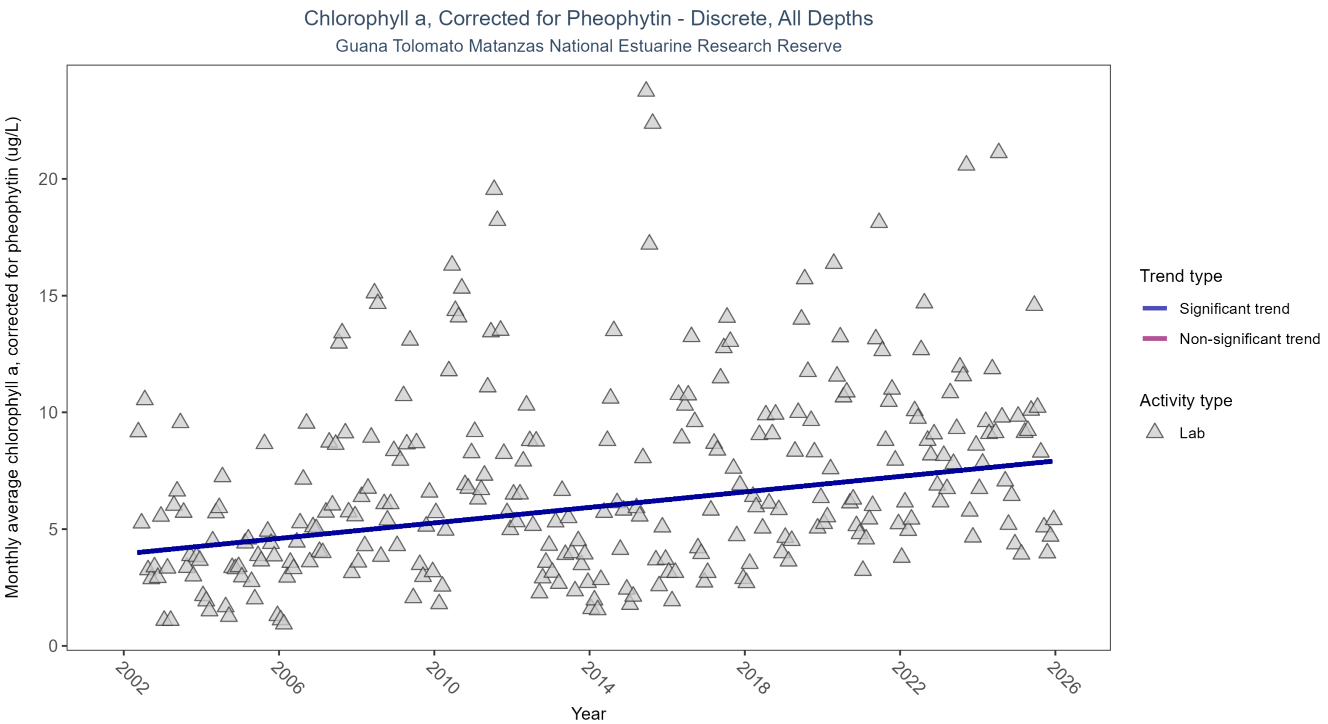

Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	8337	24	2002 - 2025	4.7	0.3	3.9382	0.1659	0

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.17 µg/L per year, indicating a decrease in water clarity.

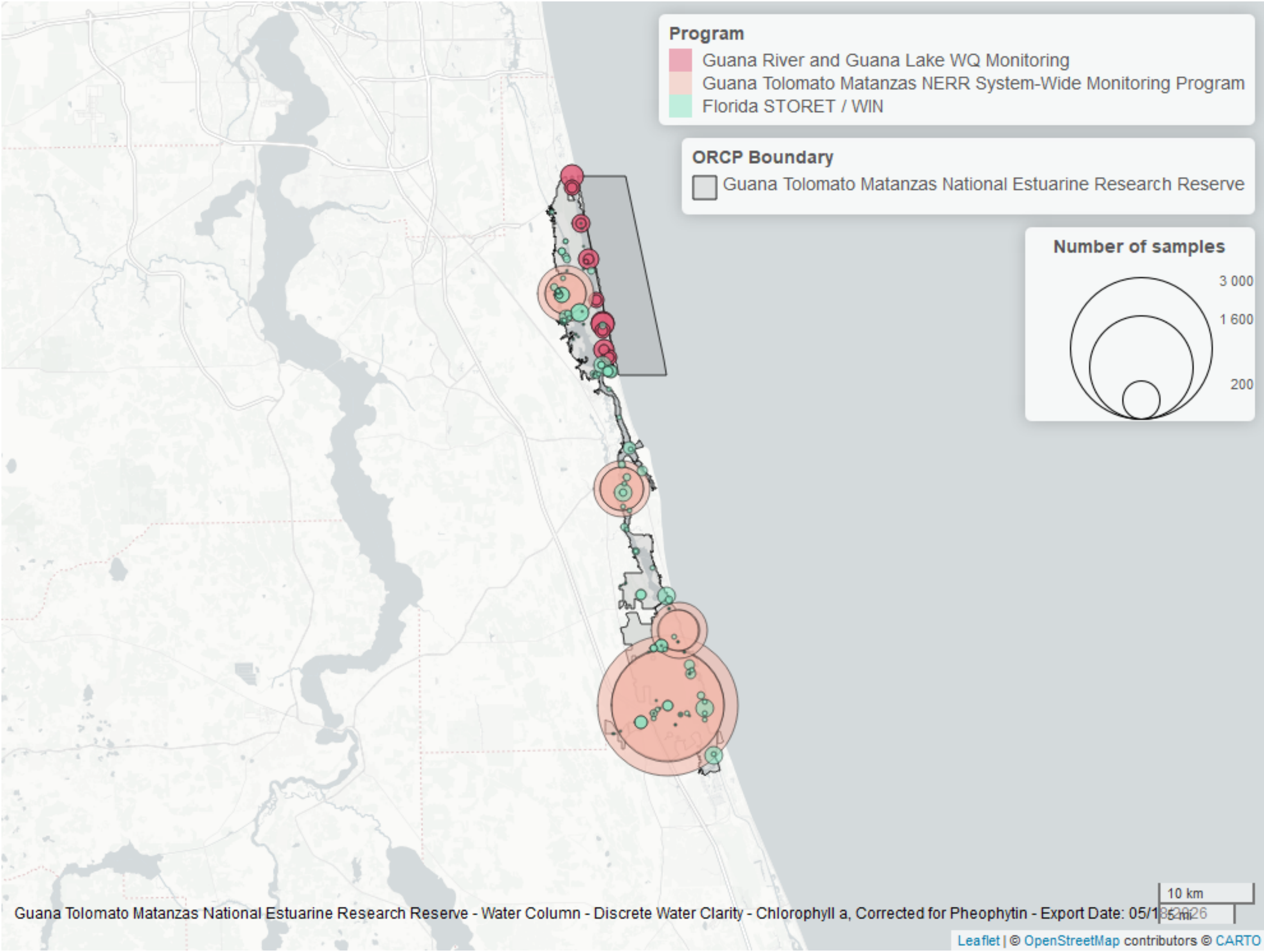

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
4054	7124	2002	2025
5002	946	2002	2025
5014	714	2017	2024

Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
5002 - Florida STORET / WIN[2]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

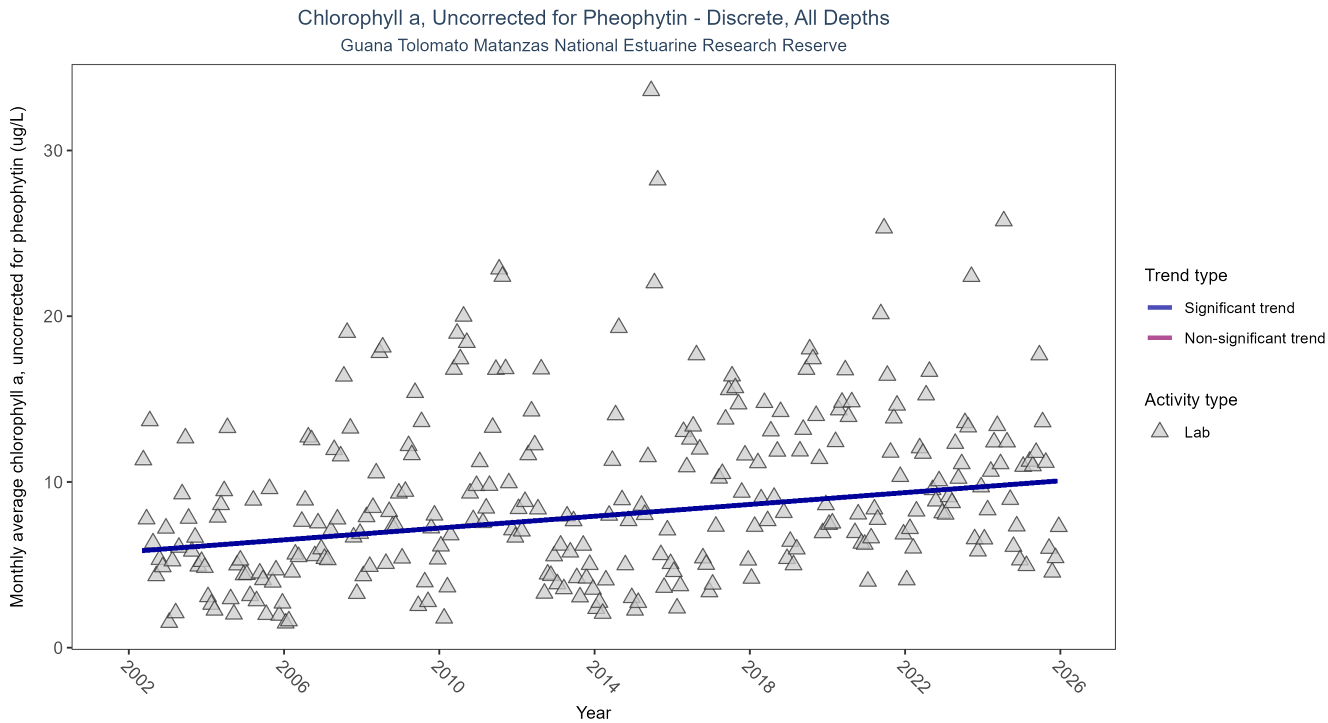

Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6544	24	2002 - 2025	6.3	0.2432	5.7922	0.1782	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.18 µg/L per year, indicating a decrease in water clarity.

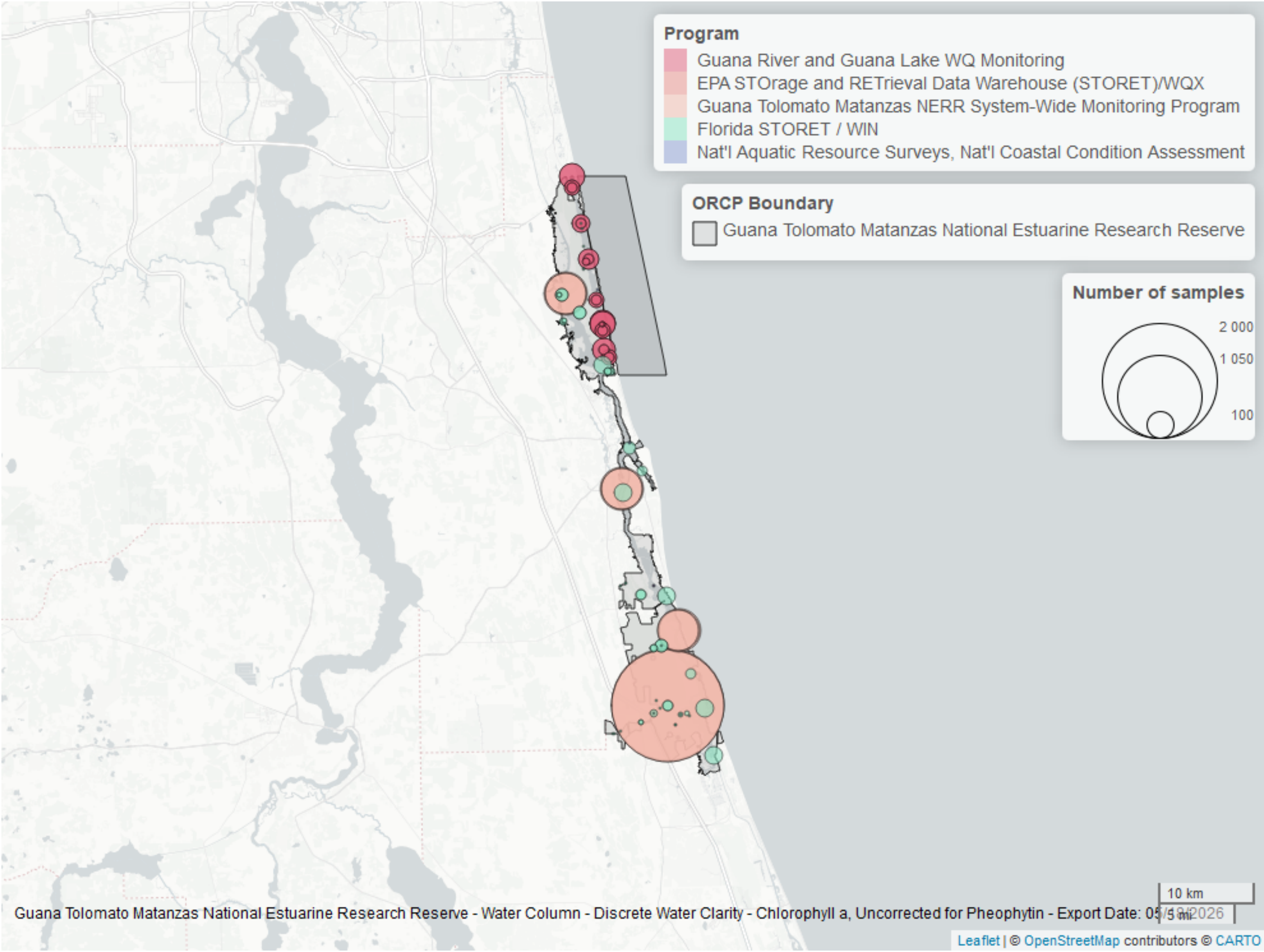

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
4054	5555	2002	2025
5014	773	2017	2024
5002	492	2008	2025
118	3	2006	2010
103	3	2006	2015

Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
5002 - Florida STORET / WIN[2]
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment[4]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

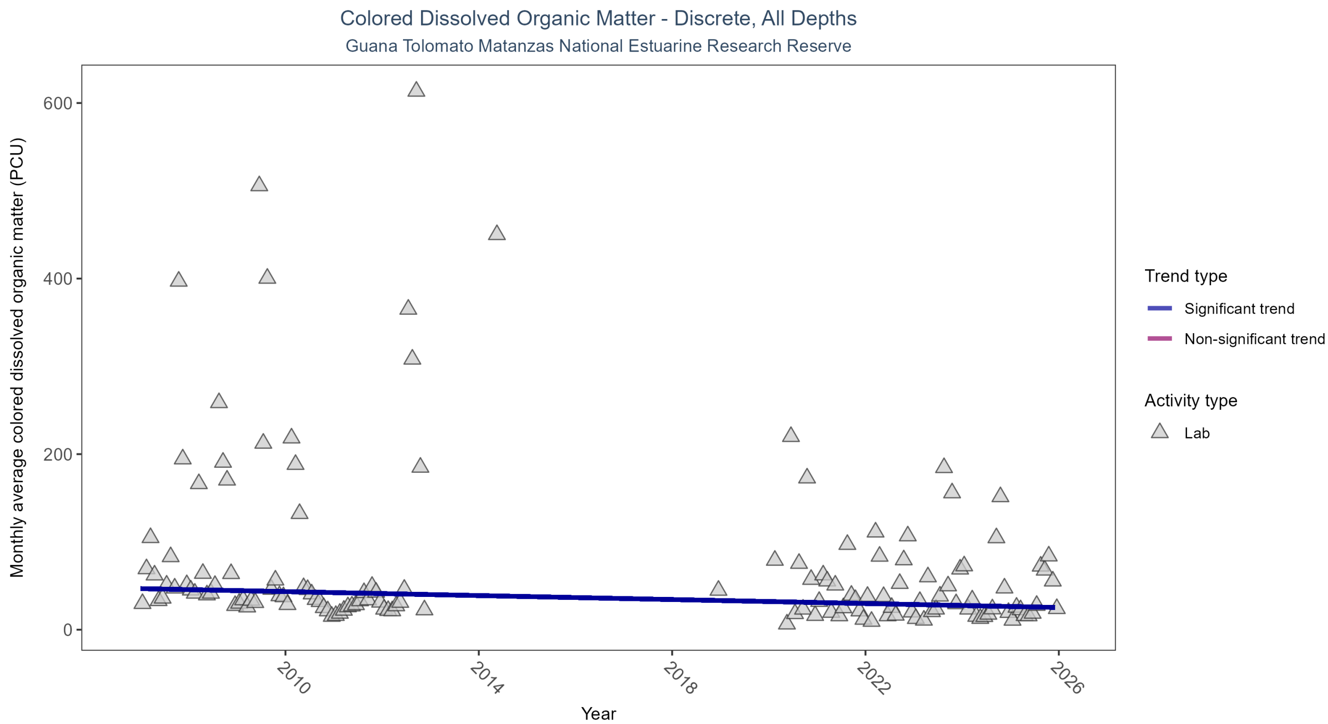

Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	1971	14	2007 - 2025	32.8	-0.2306	46.7947	-1.131	0.0004

Monthly average colored dissolved organic matter decreased by 1.13 PCU per year, indicating an increase in water clarity.

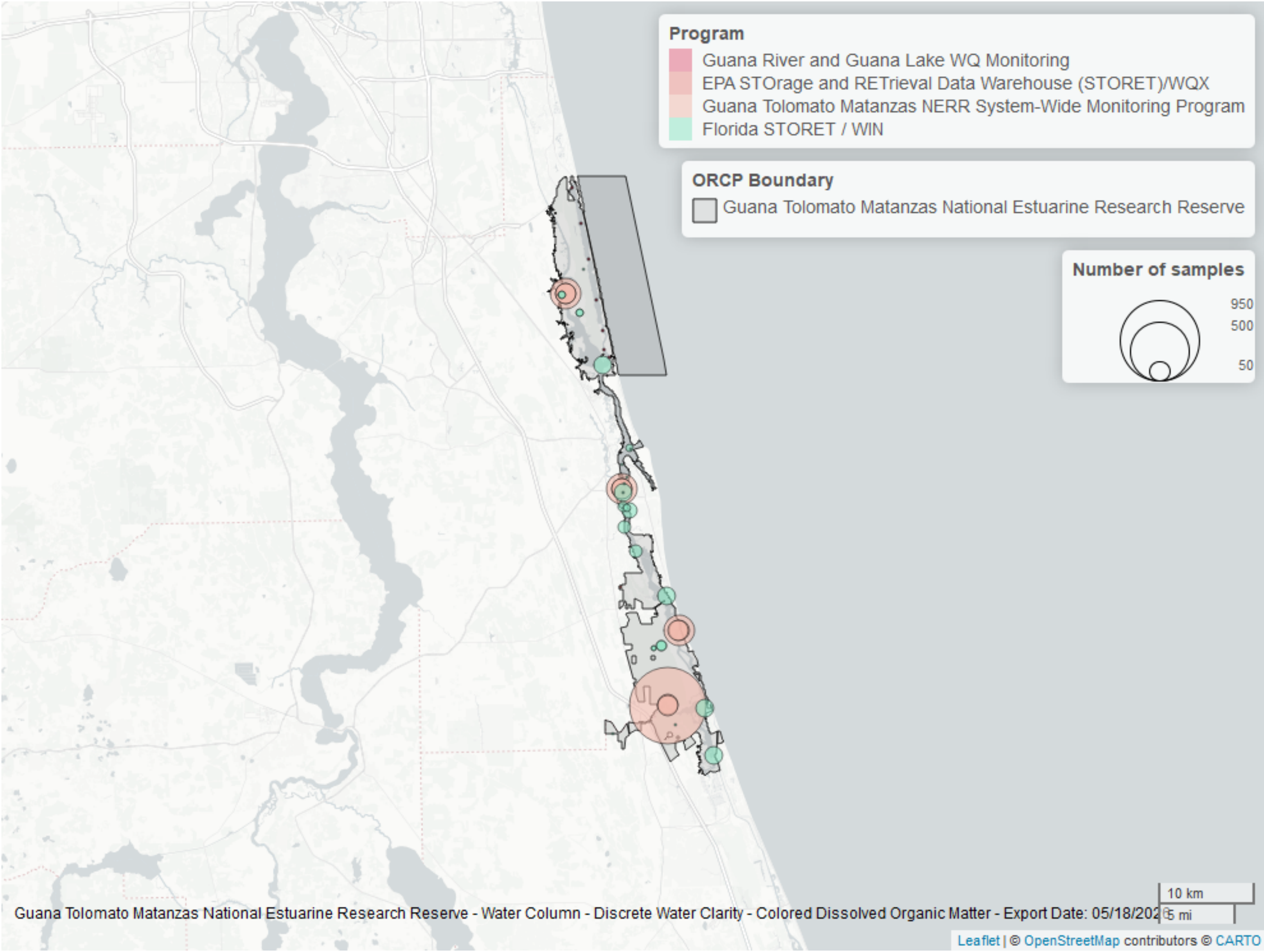

Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
4054	1566	2007	2025
5002	405	2020	2025
5014	7	2018	2018
103	2	2009	2014

Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
5002 - Florida STORET / WIN[2]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

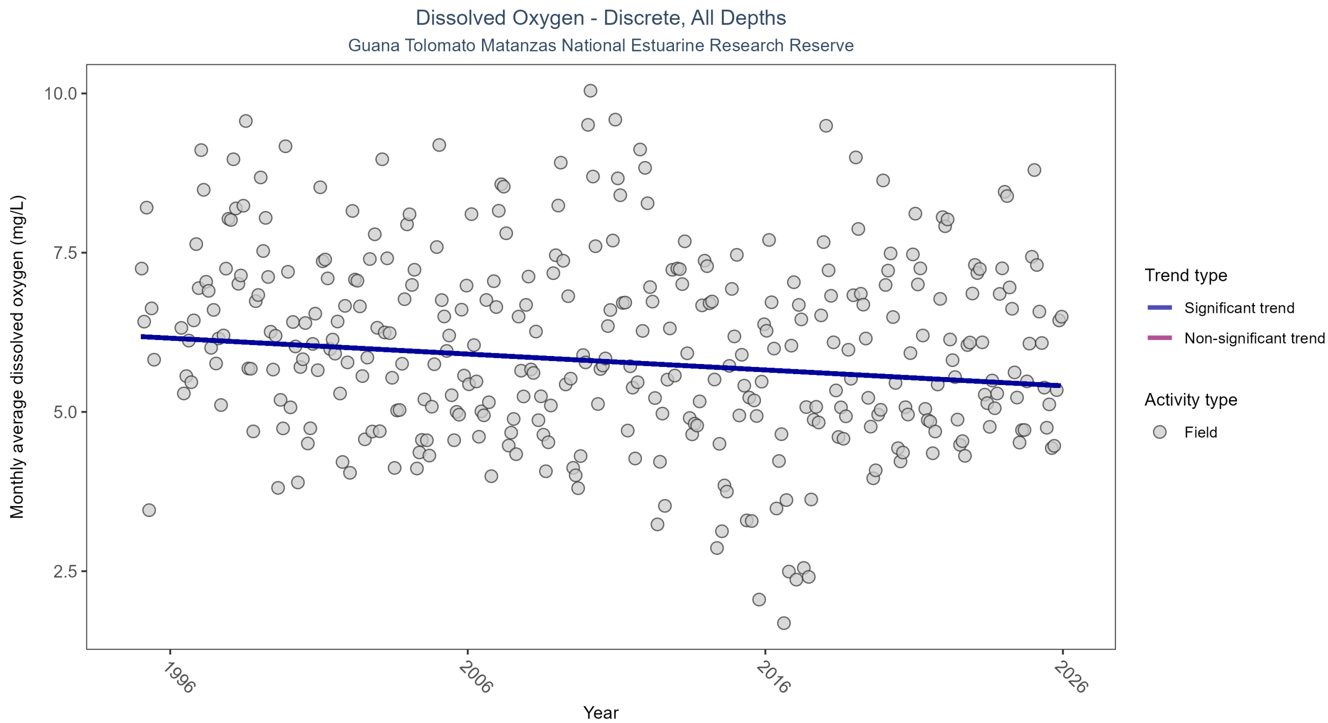

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	22781	31	1995 - 2025	6	-0.1678	6.1829	-0.0249	0

Monthly average dissolved oxygen decreased by 0.02 mg/L per year.

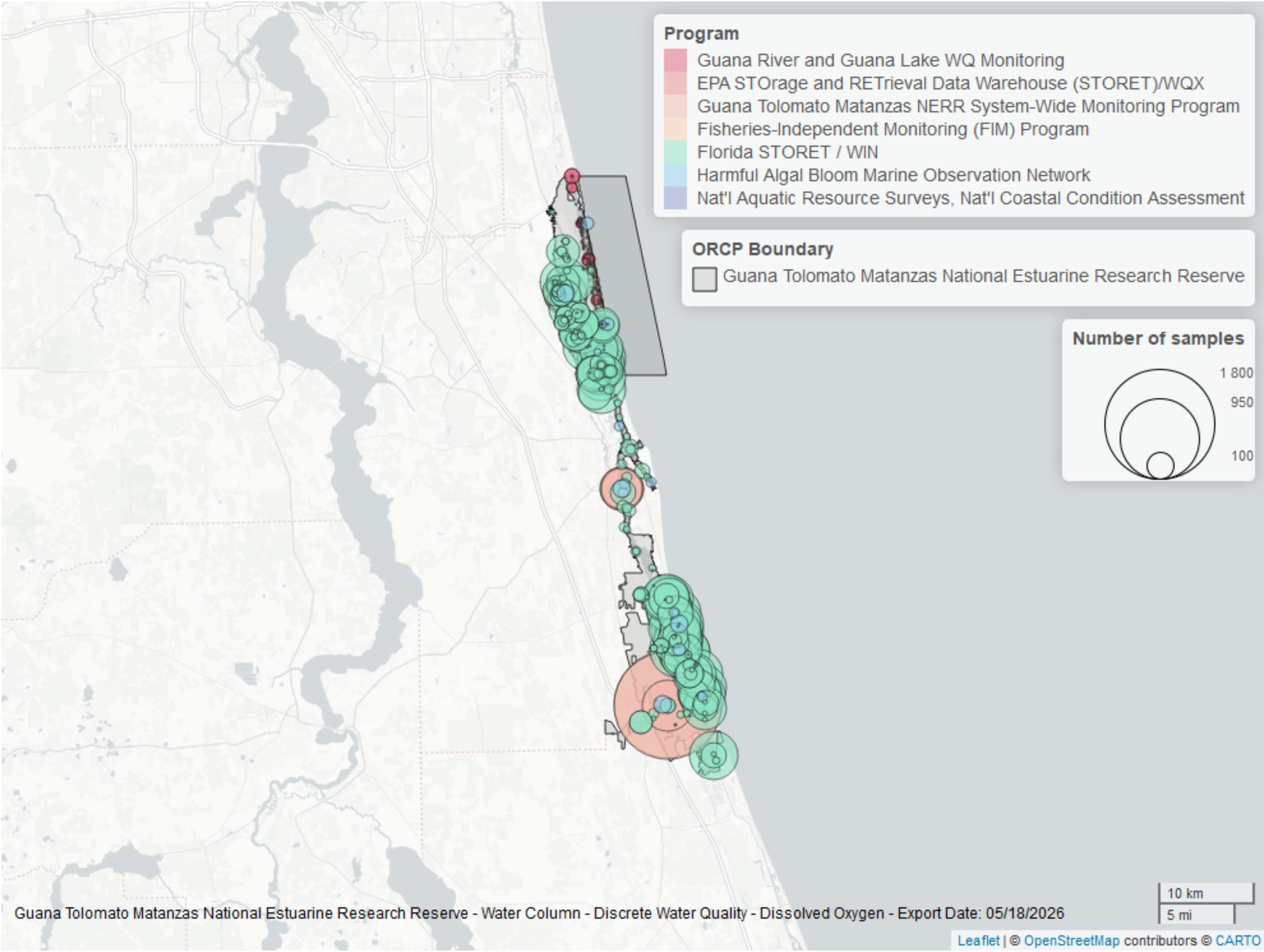

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	18375	1995	2025
4054	3747	2002	2025
95	400	2007	2018
5014	328	2017	2024
69	185	2001	2010
118	2	2006	2015
103	2	2014	2015

Program names:

5002 - Florida STORET / WIN[2]
4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
95 - Harmful Algal Bloom Marine Observation Network[6]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
69 - Fisheries-Independent Monitoring (FIM) Program[7]

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment[4]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

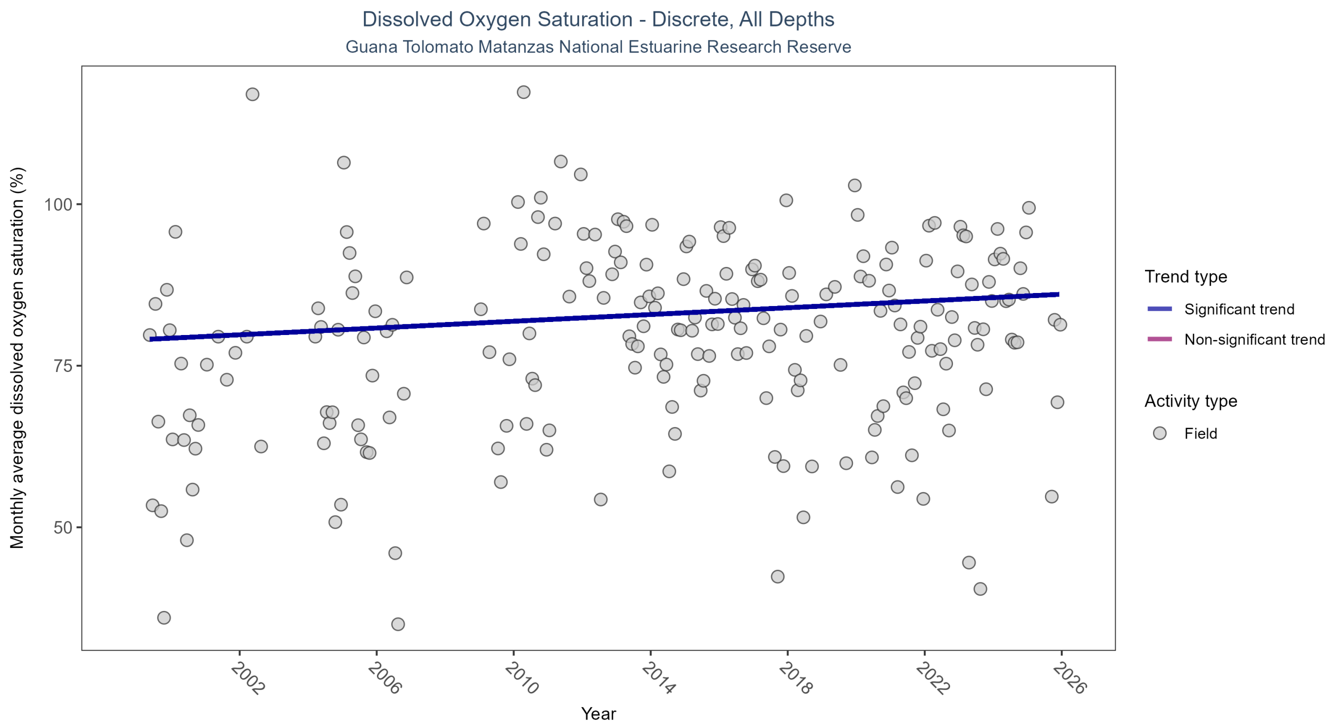

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	2522	24	1999 - 2025	81.7	0.1188	79.0065	0.2613	0.0151

Monthly average dissolved oxygen saturation increased by 0.26% per year.

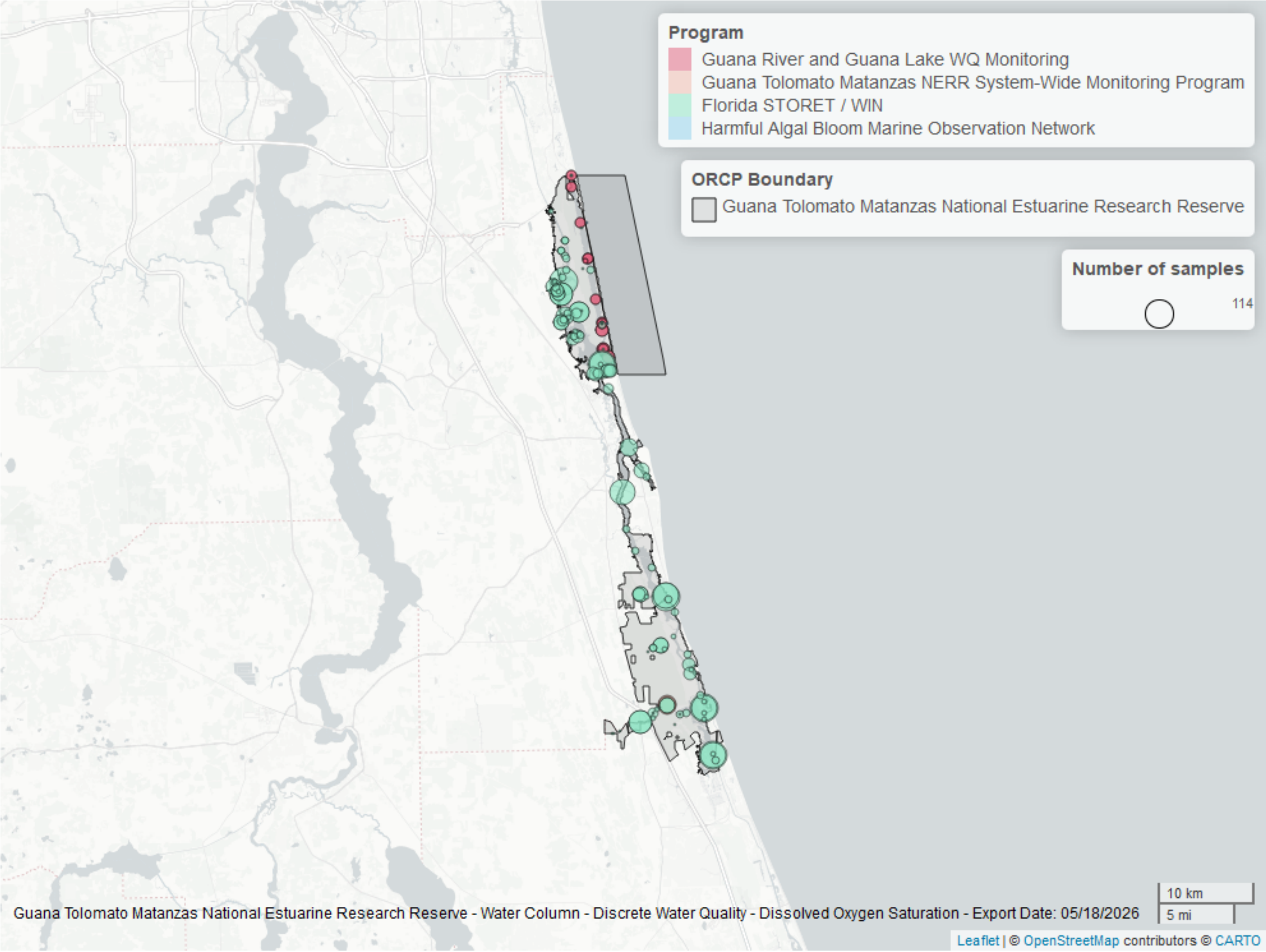

Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
5002	2279	1999	2025
5014	238	2017	2022
4054	44	2021	2023
95	3	2012	2013

Program names:

5002 - Florida STORET / WIN[2]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
95 - Harmful Algal Bloom Marine Observation Network[6]

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

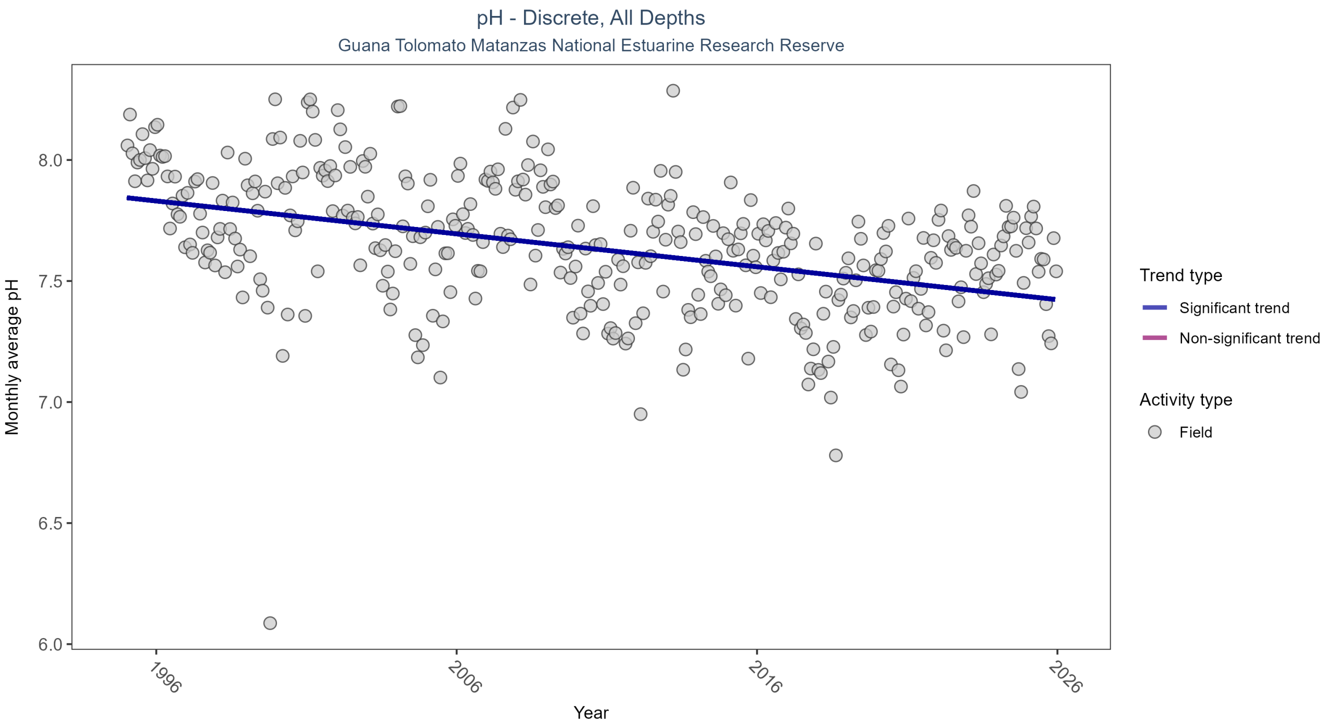

Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	19050	31	1995 - 2025	7.8	-0.3663	7.8442	-0.0136	0

Monthly average pH decreased by 0.01 pH units per year.

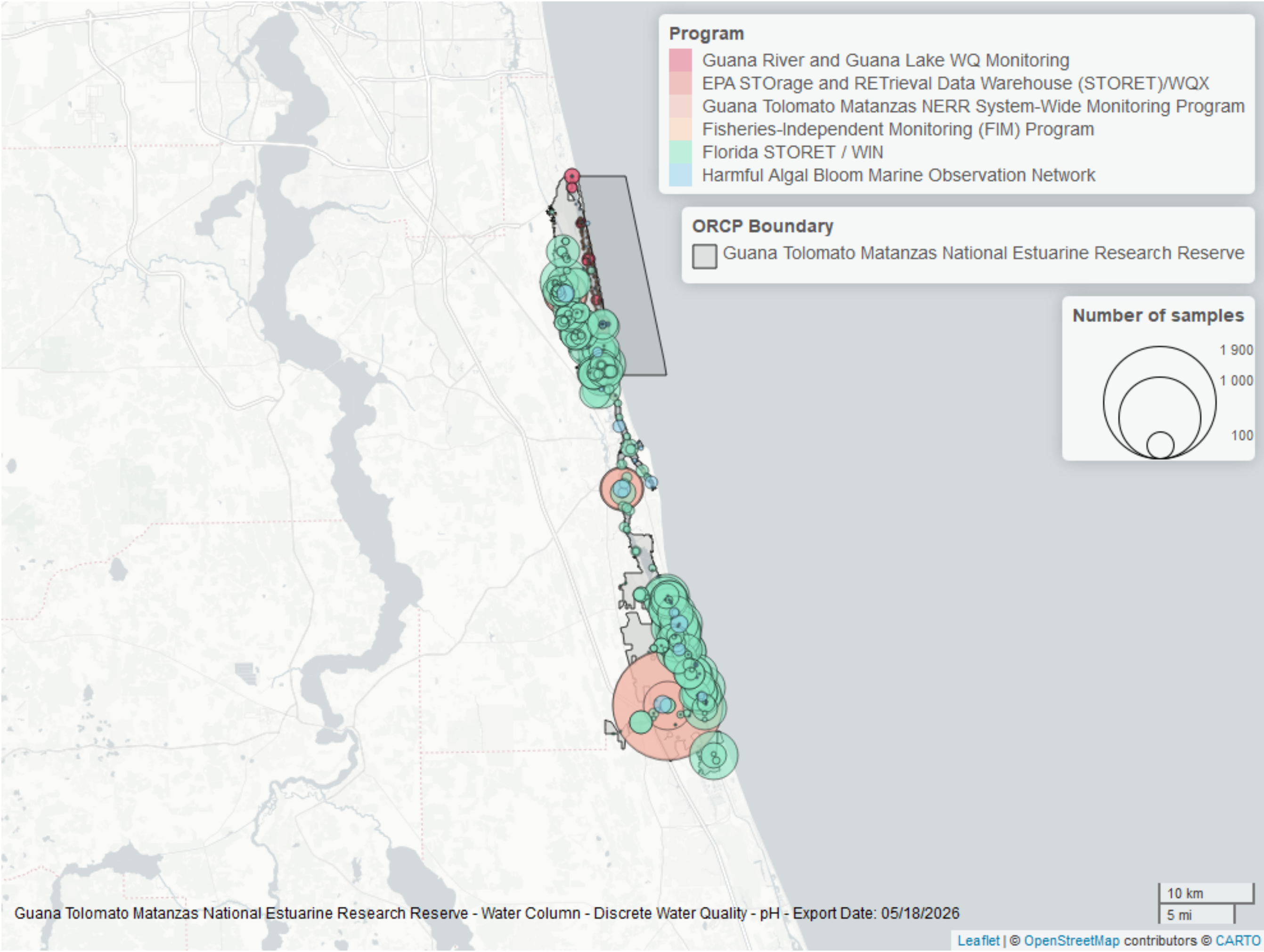

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	14992	1995	2025
4054	3781	2002	2025
95	401	2007	2018
5014	335	2017	2024
69	190	2001	2010
103	4	2014	2015

Program names:

5002 - Florida STORET / WIN[2]
4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
95 - Harmful Algal Bloom Marine Observation Network[6]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
69 - Fisheries-Independent Monitoring (FIM) Program[7]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

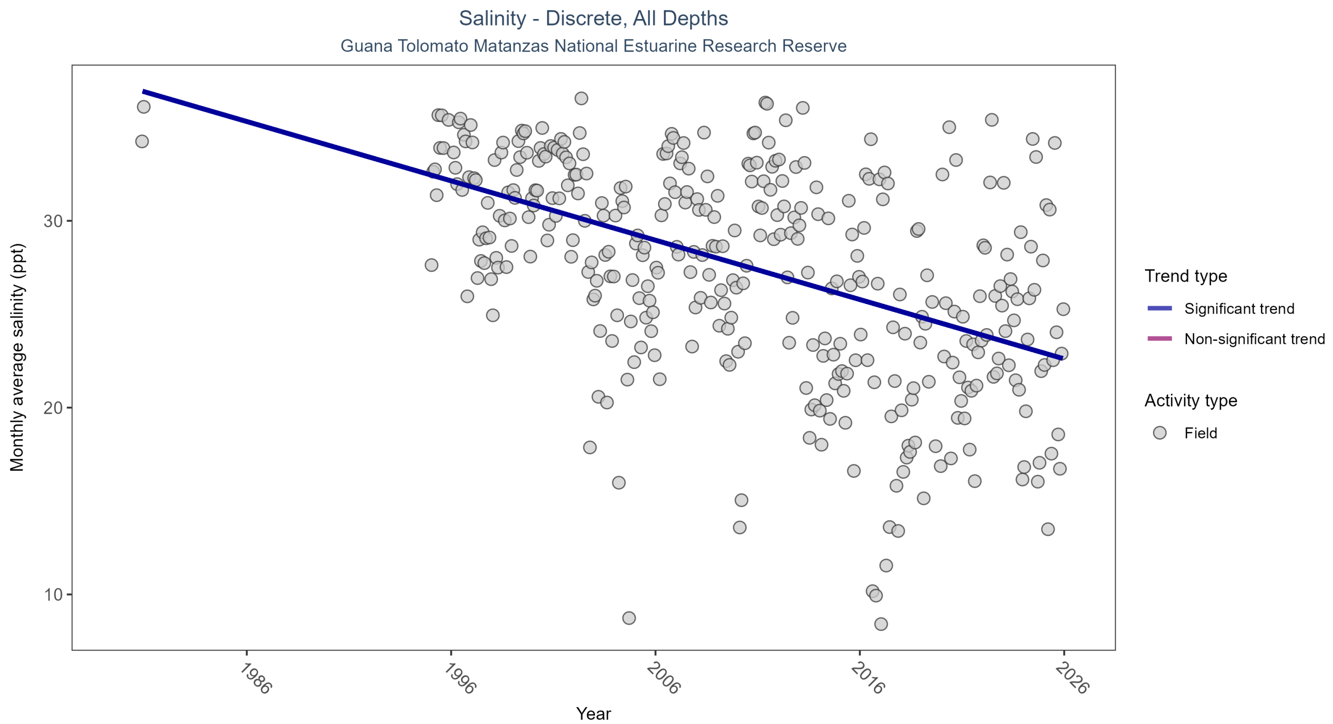

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	26179	32	1980 - 2025	31.7	-0.3675	37.2215	-0.3177	0

Monthly average salinity decreased by 0.32 ppt per year.

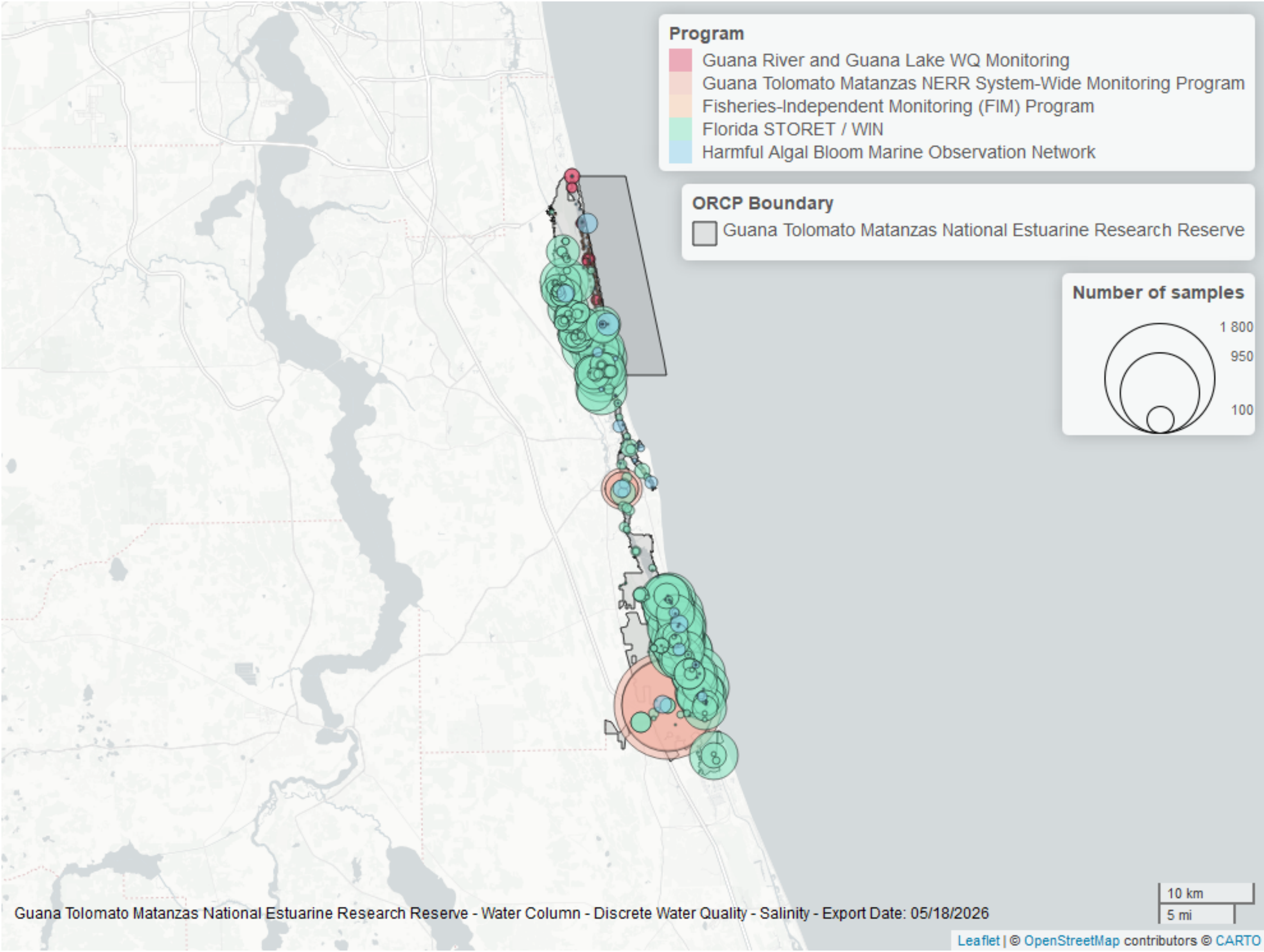

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	21034	1995	2025
4054	4366	2002	2025
95	563	1980	2018
5014	335	2017	2024
69	190	2001	2010

Program names:

5002 - Florida STORET / WIN[2]
4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
95 - Harmful Algal Bloom Marine Observation Network[6]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
69 - Fisheries-Independent Monitoring (FIM) Program[7]

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

1) TN = TKN + NO3O2;
2) TN = TKN + NO3 + NO2;
3) TN = ORGN + NH4 + NO3O2;
4) TN = ORGN + NH4 + NO2 + NO3;
5) TN = TKN + NO3;
6) TN = ORGN + NH4 + NO3;

Additional Information:

- Rules for use of sample fraction:
  - Florida Department of Environmental Protection (FDEP) report that if both "Total" and "Dissolved" components are reported, only "Total" is used. If the total is not reported, then the dissolved components are used as a best available replacement.
  - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
  - ParameterName = "Total Nitrogen"
  - SEACAR_QAQCFlagCode = "1Q"
  - SEACAR_QAQC_Description = "SEACAR Calculated"

Seasonal Kendall-Tau Trend Analysis

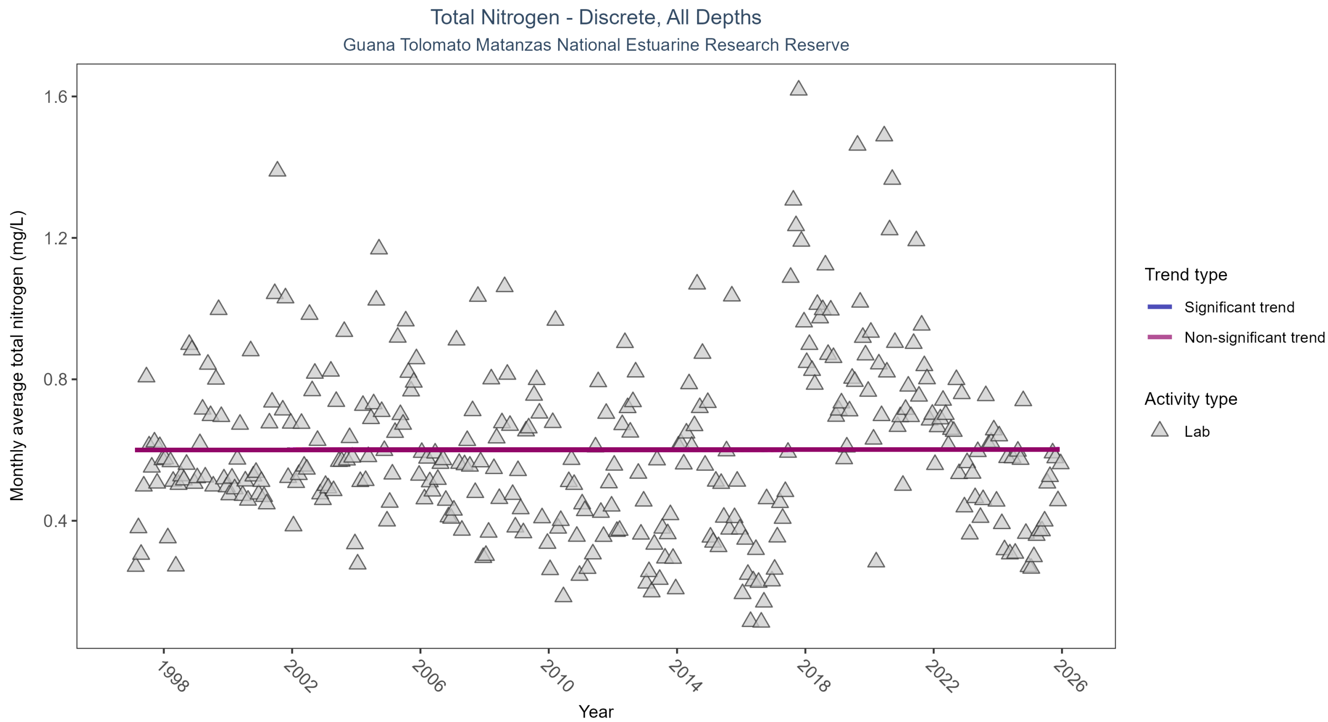

Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6124	29	1997 - 2025	0.547	0.0025	0.6002	0	0.9559

Total nitrogen showed no detectable trend between 1997 and 2025.

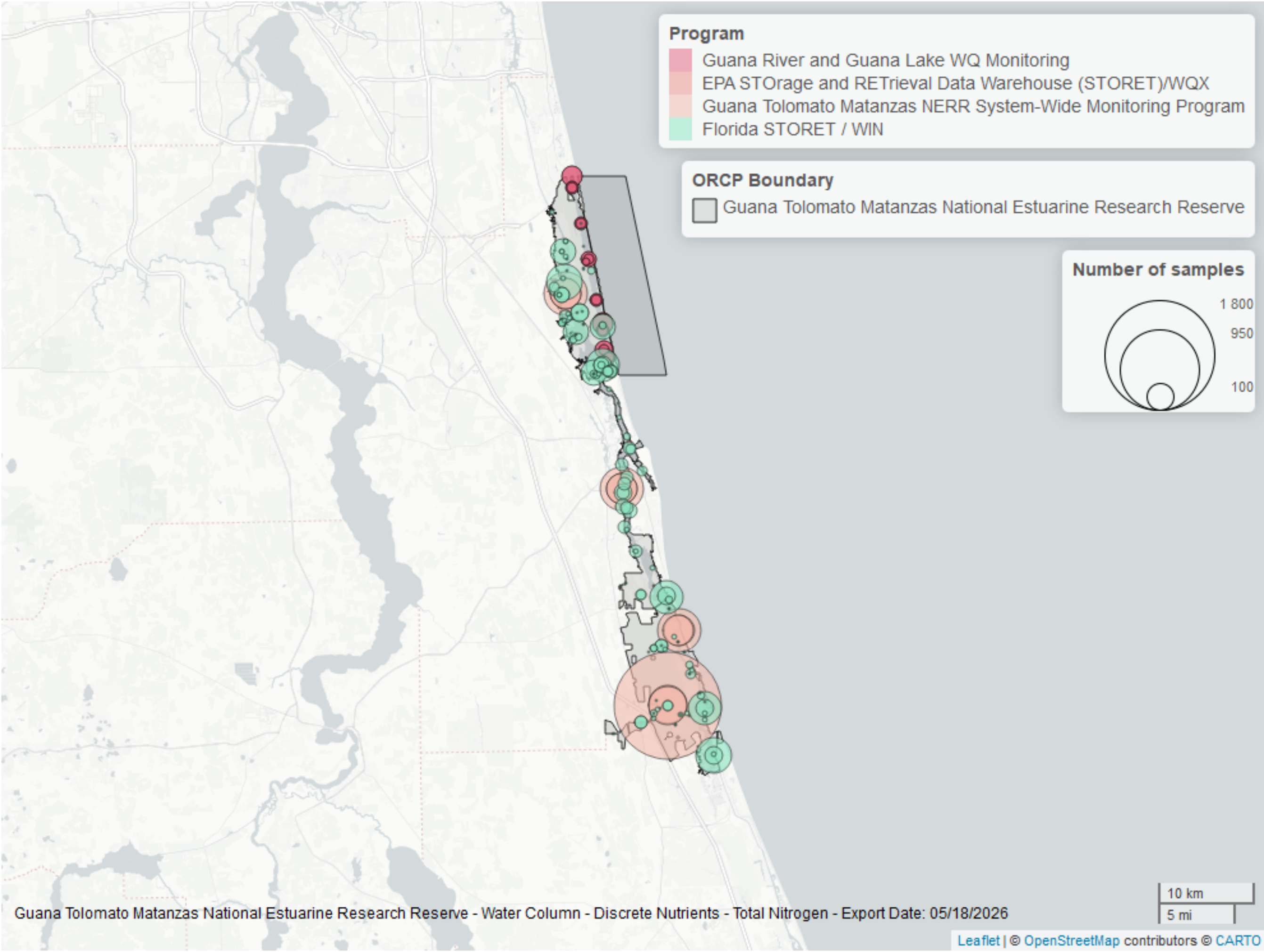

Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
4054	3297	2002	2025
5002	2465	1997	2025
5014	537	2017	2022
103	7	2006	2015

Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
5002 - Florida STORET / WIN[2]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

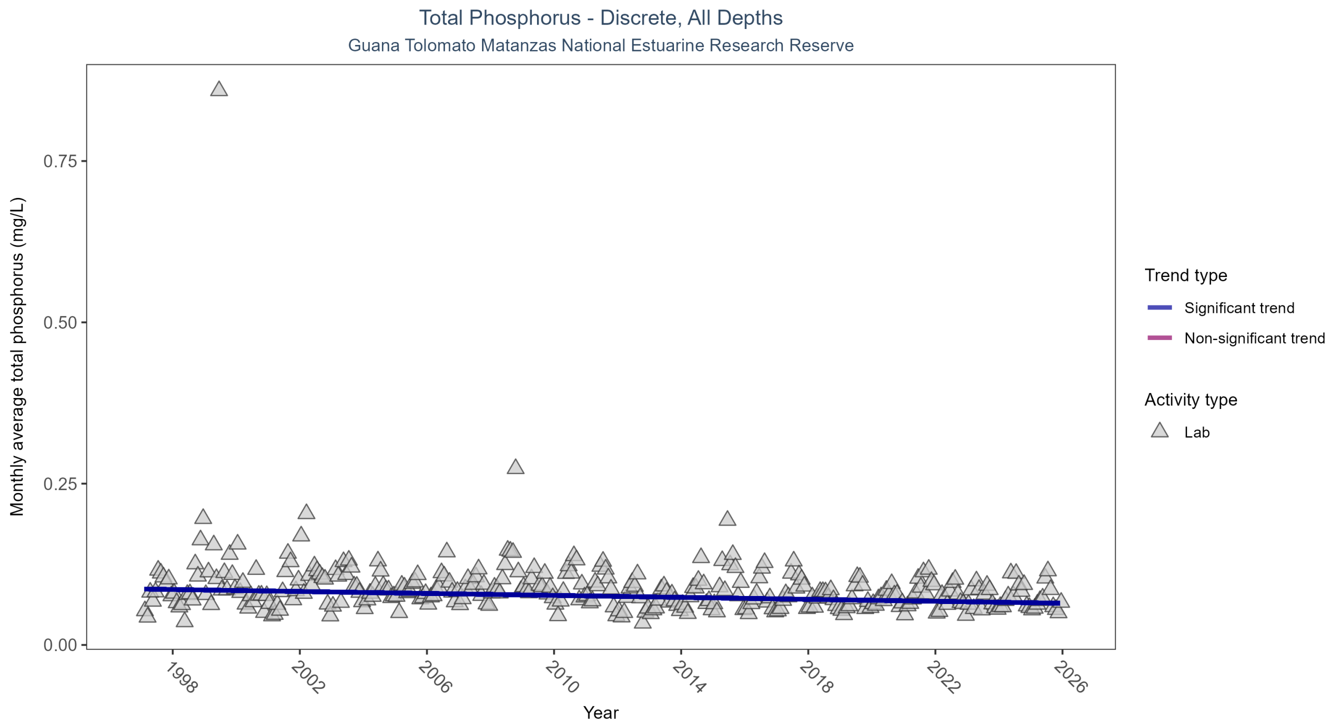

Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	9777	29	1997 - 2025	0.071	-0.2182	0.0867	-0.0008	0

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

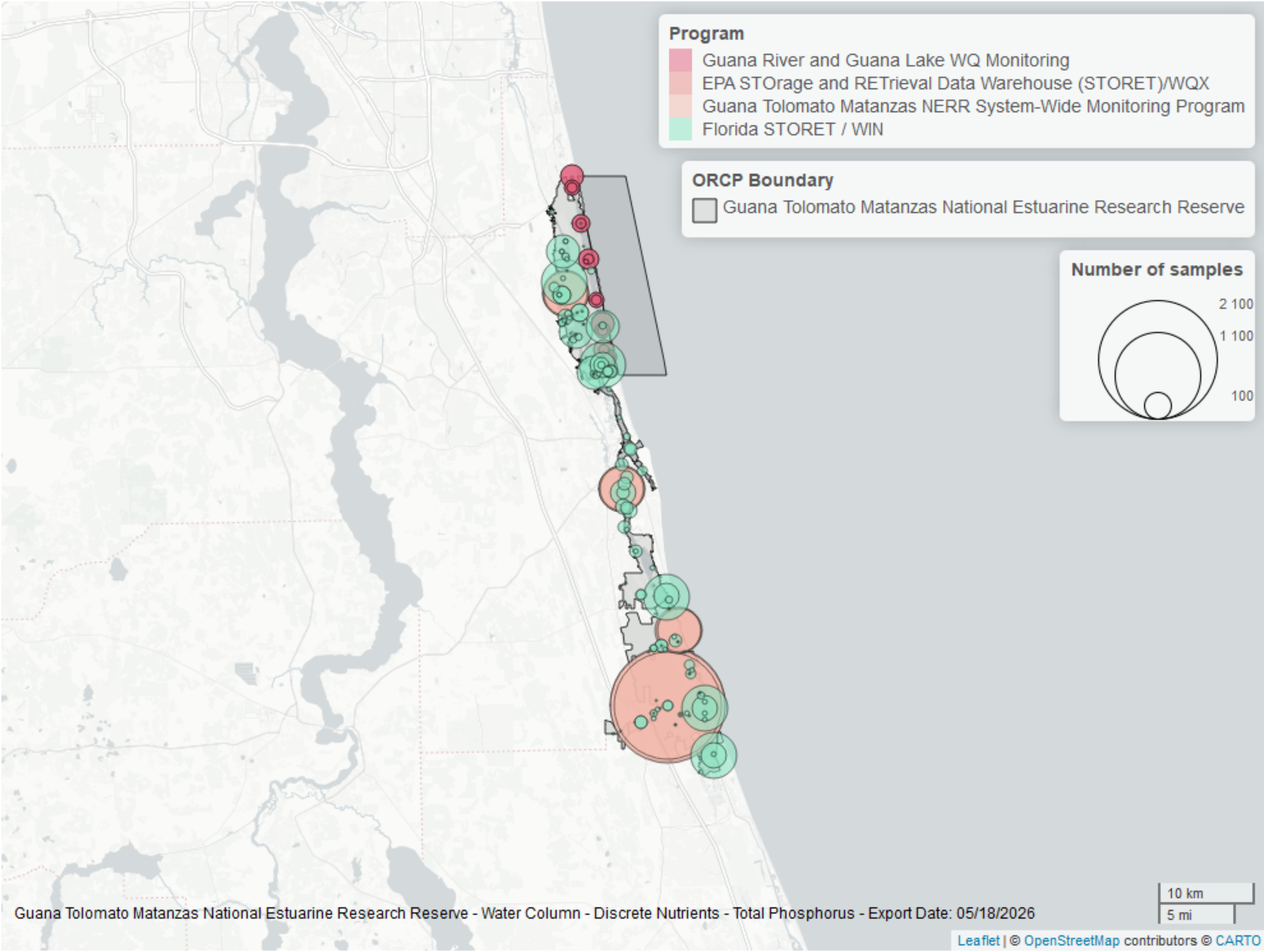

Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
4054	5654	2002	2025
5002	3735	1997	2025
5014	715	2017	2024
103	6	2006	2015

Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
5002 - Florida STORET / WIN[2]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

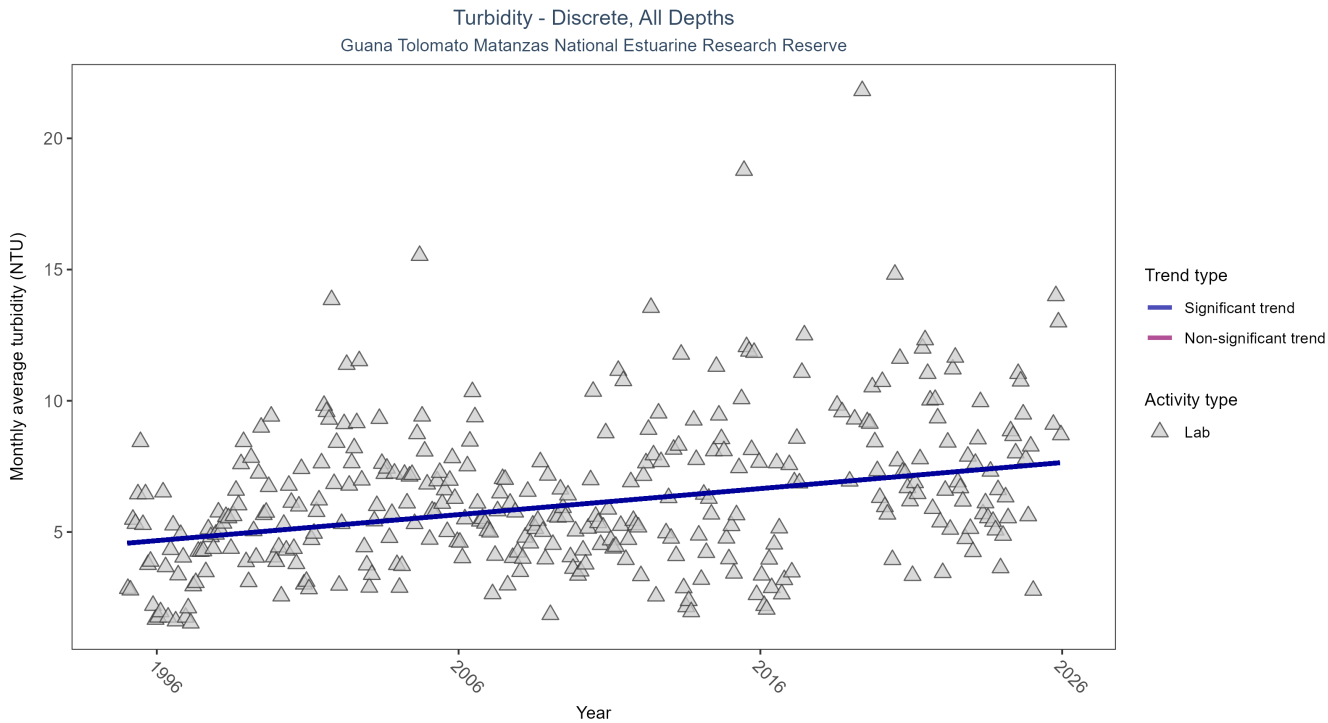

Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	14885	31	1995 - 2025	4.5	0.2656	4.572	0.0992	0

Monthly average turbidity increased by 0.1 NTU per year, indicating a decrease in water clarity.

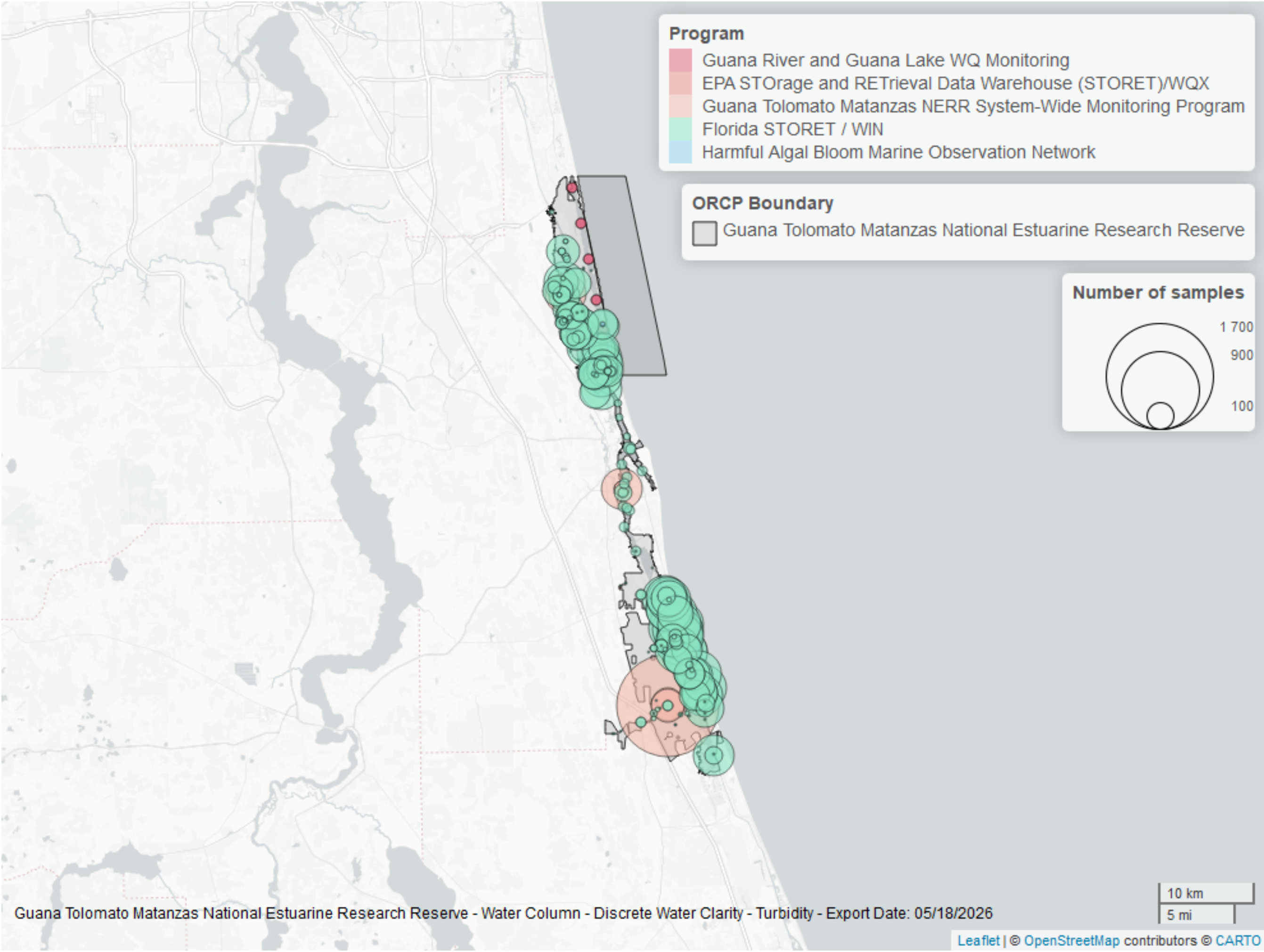

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	14745	1995	2025
4054	2683	2002	2021
5014	126	2018	2022
103	4	2006	2014
95	4	2012	2012

Program names:

5002 - Florida STORET / WIN[2]
4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]
95 - Harmful Algal Bloom Marine Observation Network[6]

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

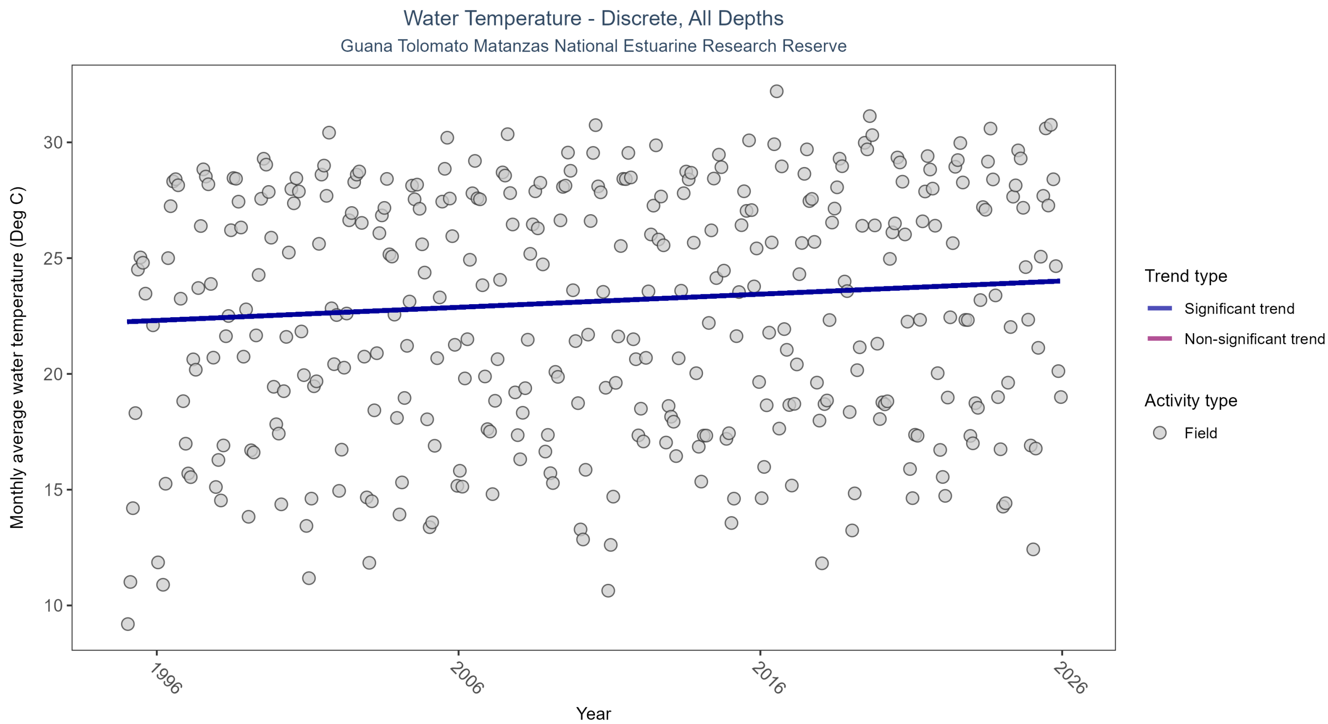

Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	25682	31	1995 - 2025	23.3	0.2292	22.2508	0.0569	0

Monthly average water temperature increased by 0.06°C per year.

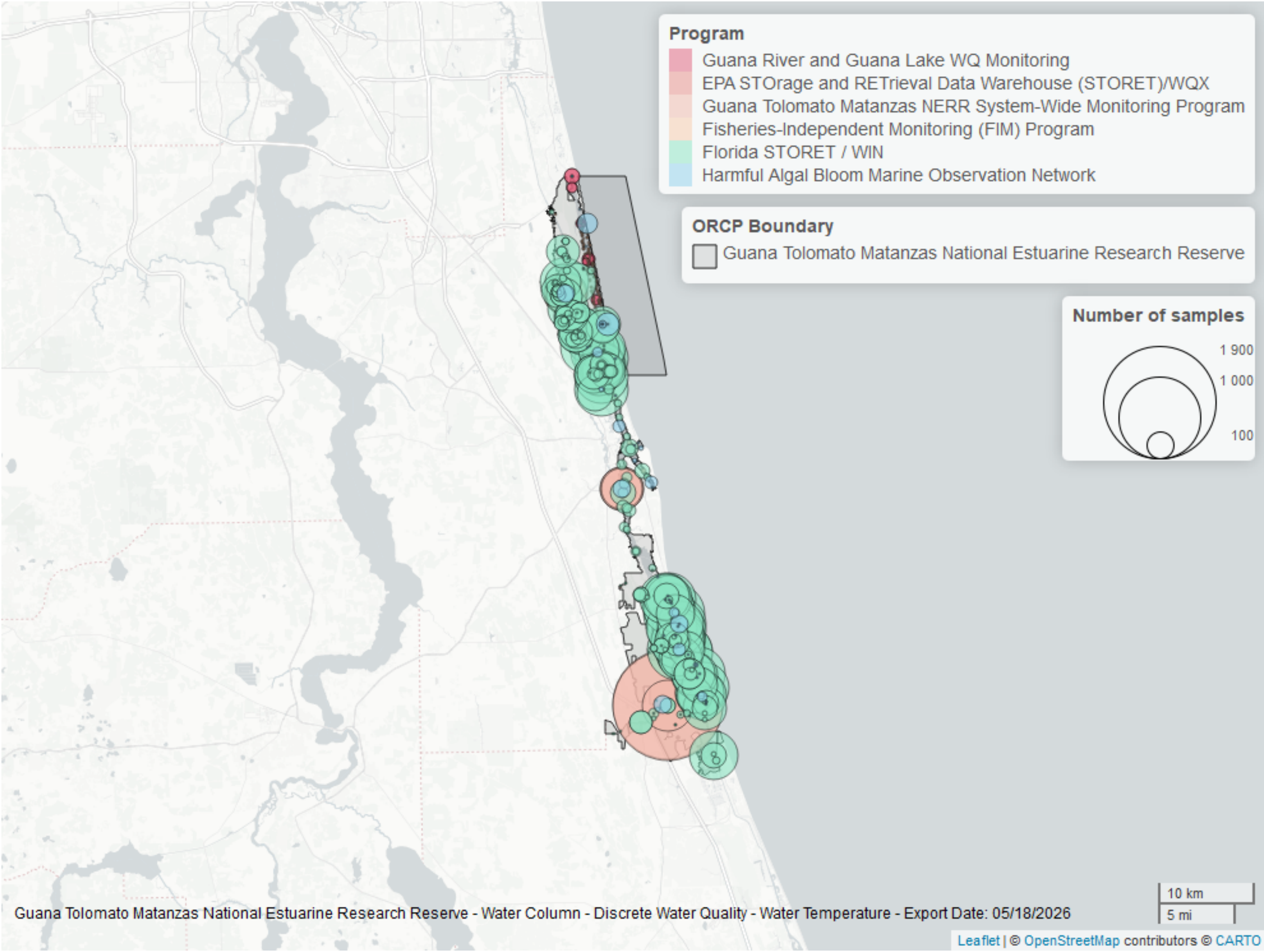

Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	21533	1995	2025
4054	3856	2002	2025
95	534	2007	2018
5014	335	2017	2024
69	190	2001	2010
103	1	2014	2014

Program names:

5002 - Florida STORET / WIN[2]
4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
95 - Harmful Algal Bloom Marine Observation Network[6]
5014 - Guana River and Guana Lake Water Quality Monitoring[3]
69 - Fisheries-Independent Monitoring (FIM) Program[7]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[5]

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Guana Tolomato Matanzas National Estuarine Research Reserve

Table 28: Station overview for Continuous parameters by Program

ProgramID	*ProgramLocationID*	*Years of Data*	*Use in Analysis*	*Parameters*
4054	gtmfmwq	26	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
4054	gtmpcwq	26	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
4054	gtmpiwq	26	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
4054	gtmsswq	25	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
5062	872-0494	2	FALSE	Water Temperature , Salinity , Specific Conductivity

Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program[1]
5062 - FDEP Bureau of Survey and Mapping Continuous Water Quality Program[8]

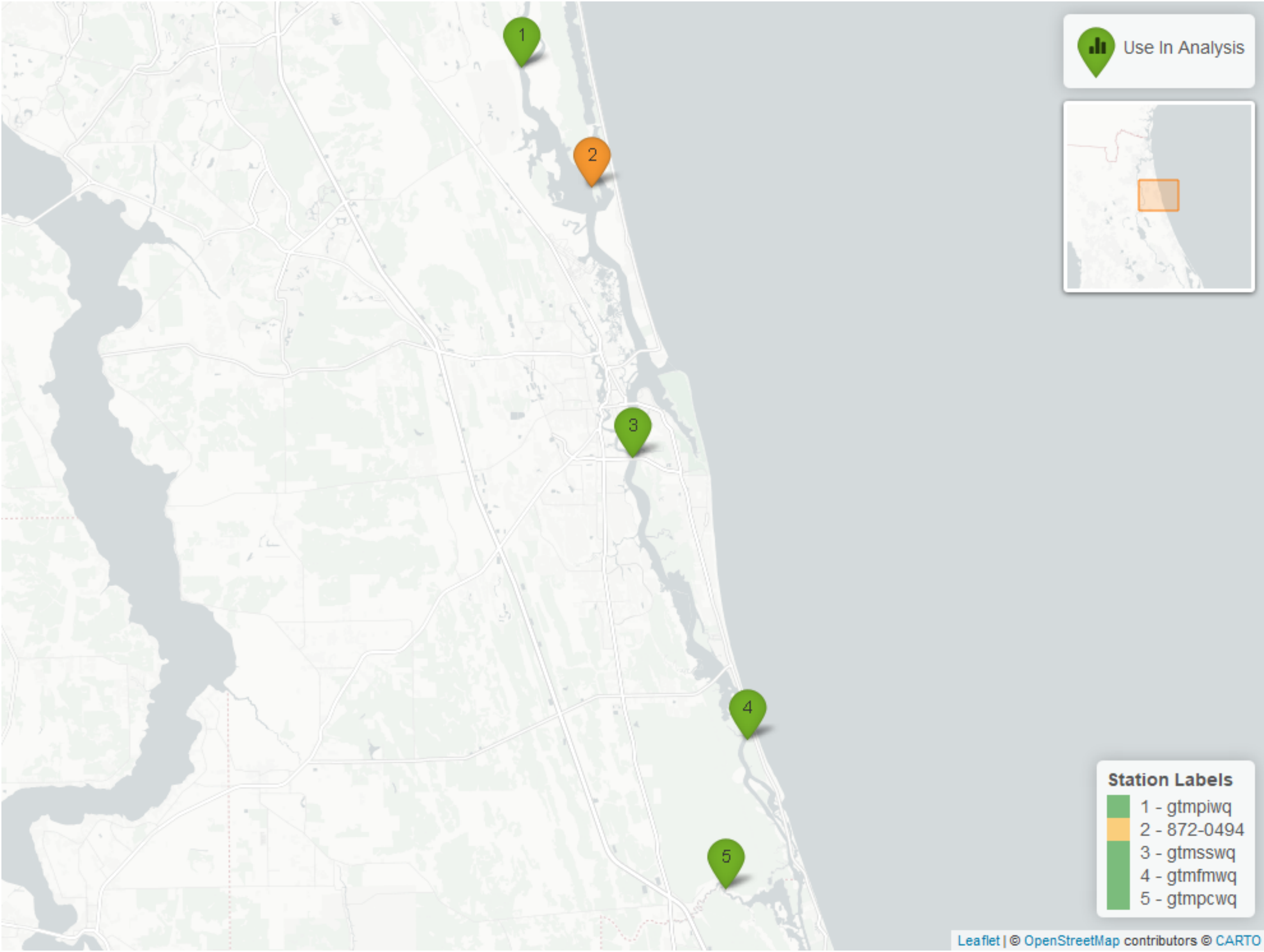

Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

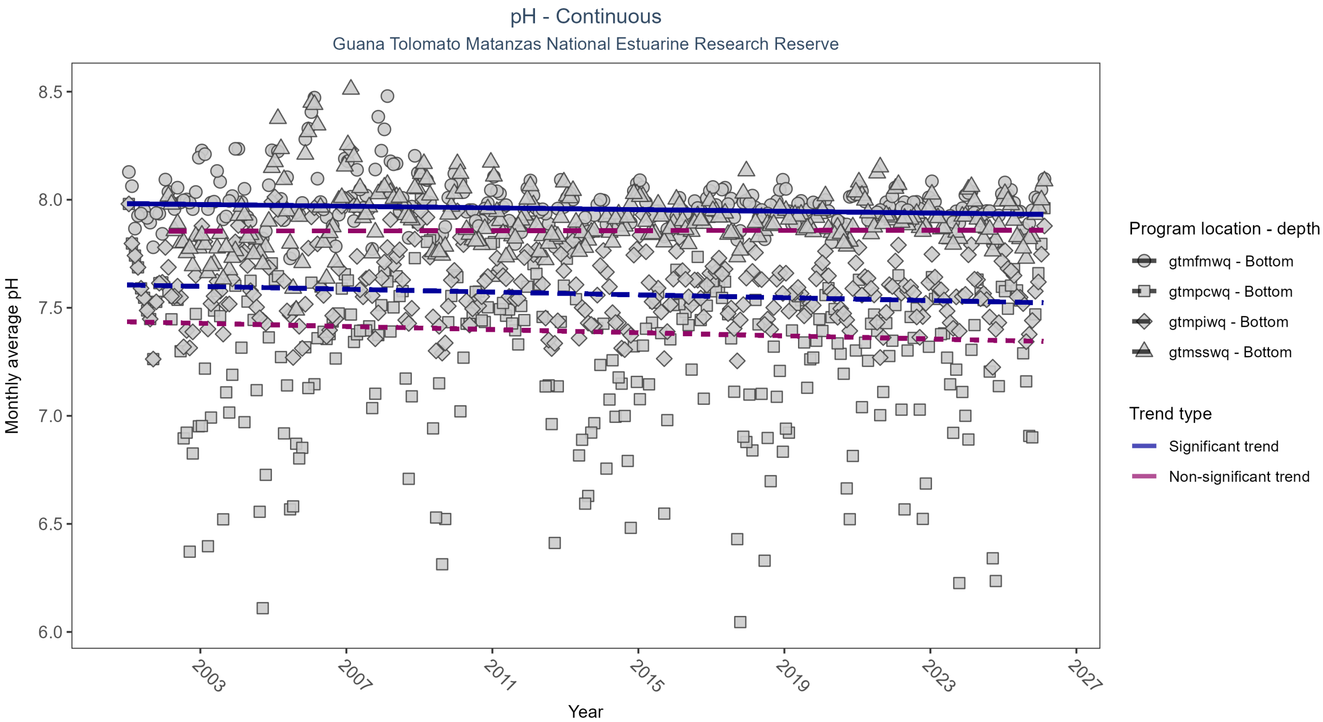

Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	696449	26	2001 - 2026	7.6	-0.17	7.61	0	0.00
gtmfmwq	Significantly decreasing trend	702168	26	2001 - 2026	8.0	-0.14	7.98	0	0.00
gtmsswq	No significant trend	674734	25	2002 - 2026	7.9	0.01	7.86	0	0.85
gtmpcwq	No significant trend	730315	26	2001 - 2026	7.4	-0.08	7.44	0	0.06

At two program locations, monthly average pH decreased by 0.00 pH units per year. No detectable change in monthly average pH was observed at two locations.

Dissolved Oxygen Saturation - Continuous

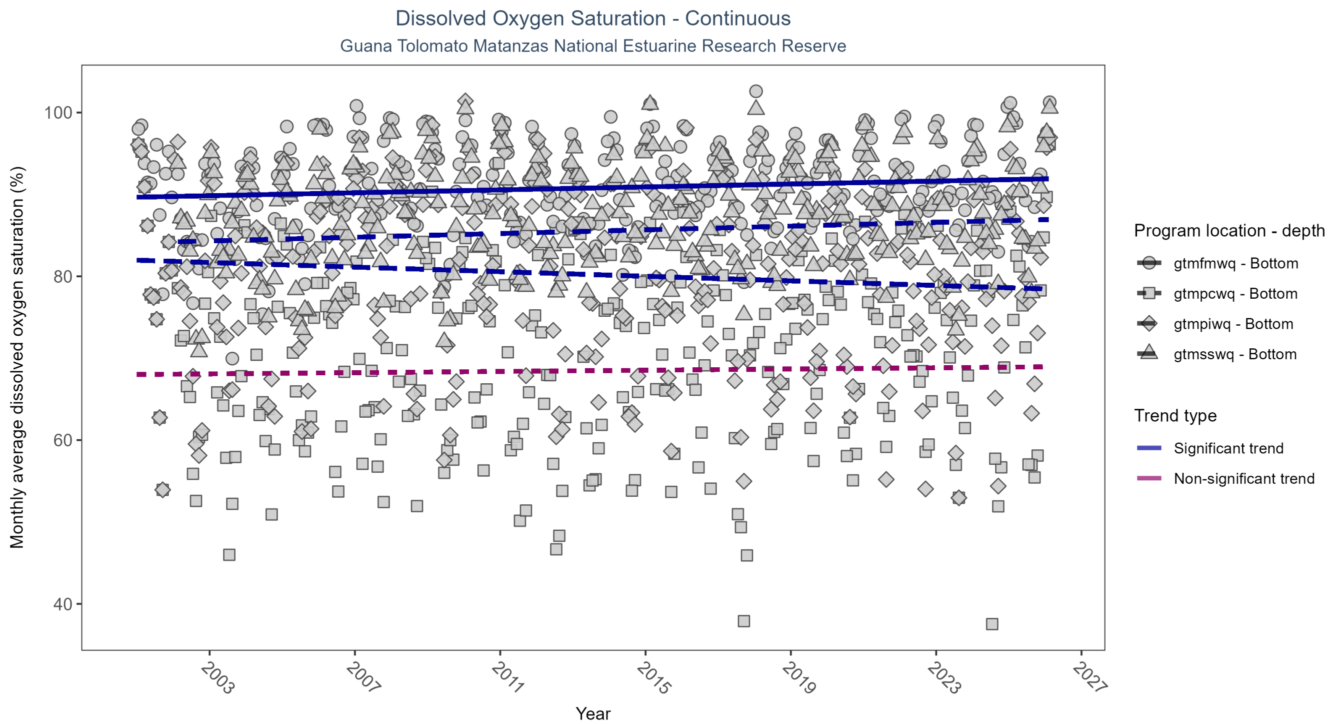

Figure 25: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	705219	26	2001 - 2026	82.2	-0.15	81.97	-0.14	0.00
gtmfmwq	Significantly increasing trend	731140	26	2001 - 2026	92.7	0.16	89.66	0.09	0.00
gtmsswq	Significantly increasing trend	704965	25	2002 - 2026	89.5	0.16	84.21	0.11	0.00
gtmpcwq	No significant trend	737581	26	2001 - 2026	71.5	0.03	68.01	0.04	0.49

At two program locations, monthly average dissolved oxygen saturation increased by 0.09% per year at one site and by 0.11% per year at the other. At one program location, monthly average dissolved oxygen saturation decreased by 0.14% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location.

Dissolved Oxygen - Continuous

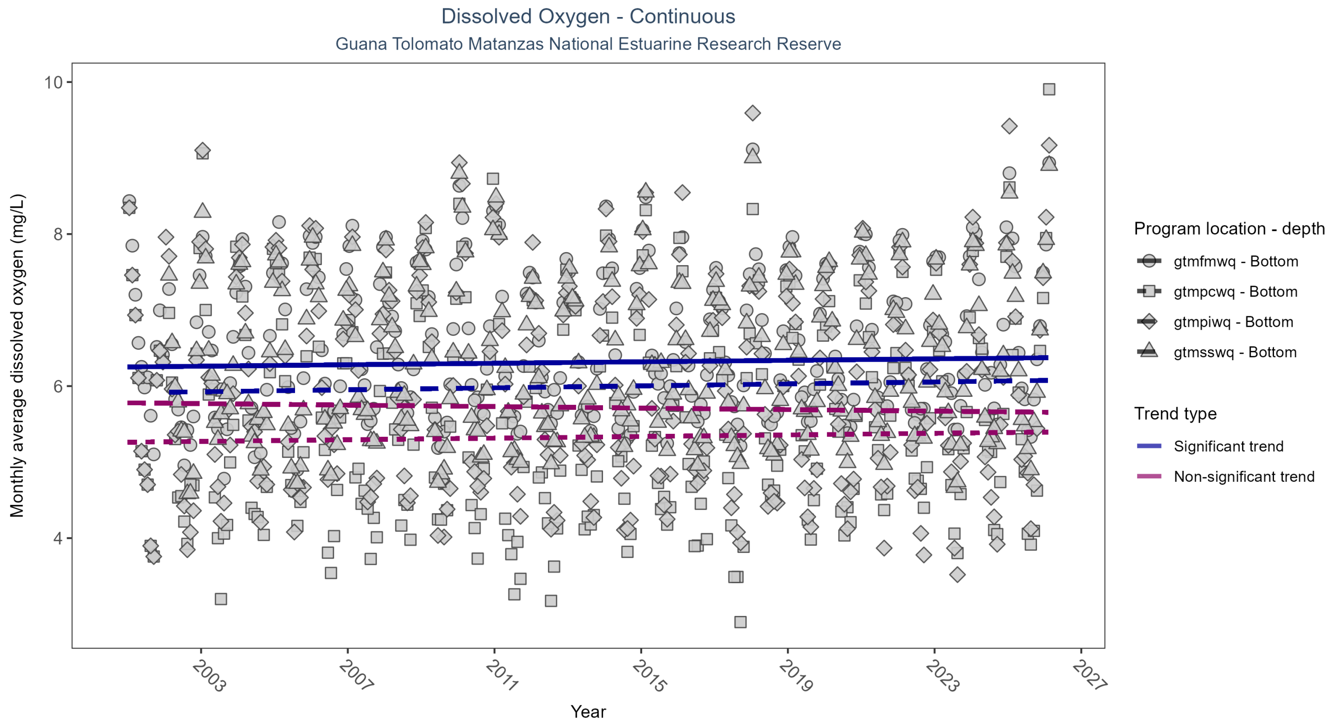

Figure 26: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	No significant trend	699401	26	2001 - 2026	5.9	-0.06	5.78	0.00	0.13
gtmfmwq	Significantly increasing trend	718097	26	2001 - 2026	6.5	0.10	6.25	0.00	0.02
gtmsswq	Significantly increasing trend	699026	25	2002 - 2026	6.3	0.10	5.92	0.01	0.02
gtmpcwq	No significant trend	736088	26	2001 - 2026	5.5	0.05	5.26	0.01	0.19

At two program locations, monthly average dissolved oxygen increased by less than 0.01 mg/L per year at one site and by 0.01 mg/L per year at the other. No detectable change in monthly average dissolved oxygen was observed at two locations.

Turbidity - Continuous

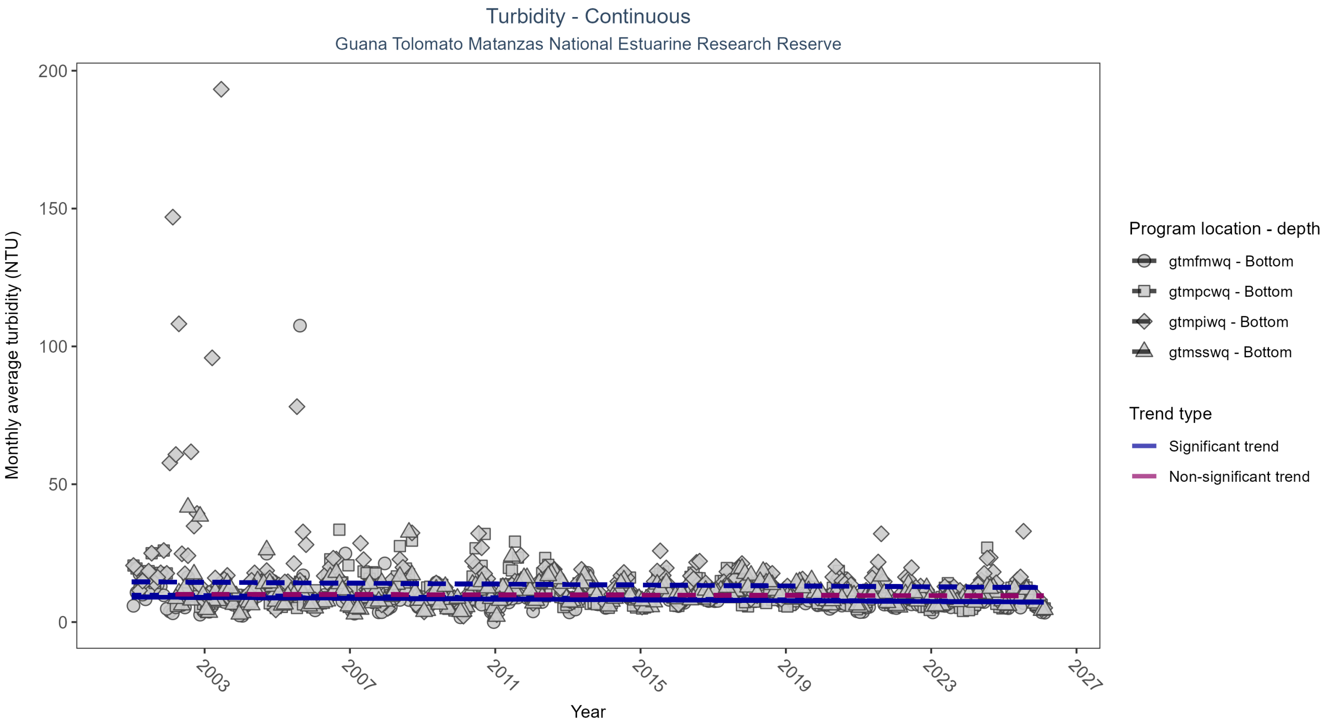

Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	679520	26	2001 - 2026	10	-0.12	14.61	-0.08	0.00
gtmfmwq	Significantly decreasing trend	717246	26	2001 - 2026	7	-0.14	9.08	-0.07	0.00
gtmsswq	No significant trend	670272	25	2002 - 2026	9	-0.04	10.11	-0.02	0.38
gtmpcwq	Significantly decreasing trend	721298	26	2001 - 2026	9	-0.17	9.80	-0.10	0.00

At three program locations, monthly average turbidity decreased between 0.07 and 0.1 NTU per year. No detectable change in monthly average turbidity was observed at one location.

Water Temperature - Continuous

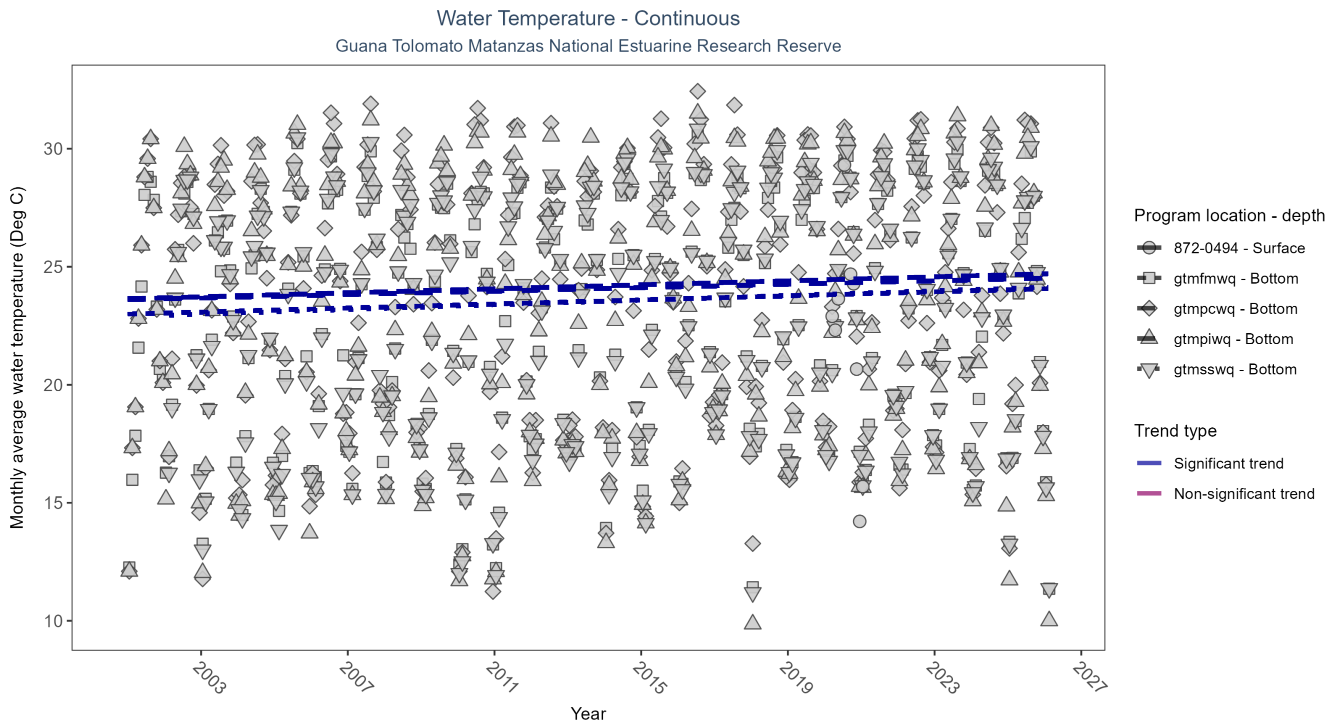

Figure 28: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly increasing trend	753811	26	2001 - 2026	24.20	0.18	23.6	0.04	0
872-0494	Insufficient data to calculate trend	35473	2	2020 - 2021	22.34	-	-	-	-
gtmfmwq	Significantly increasing trend	751023	26	2001 - 2026	23.70	0.21	22.99	0.04	0
gtmsswq	Significantly increasing trend	717305	25	2002 - 2026	23.80	0.22	22.94	0.05	0
gtmpcwq	Significantly increasing trend	754434	26	2001 - 2026	24.30	0.15	23.64	0.04	0

At four program locations, monthly average water temperature increased between 0.04 and 0.05°C per year. There was insufficient data to fit a model for one location.

Salinity - Continuous

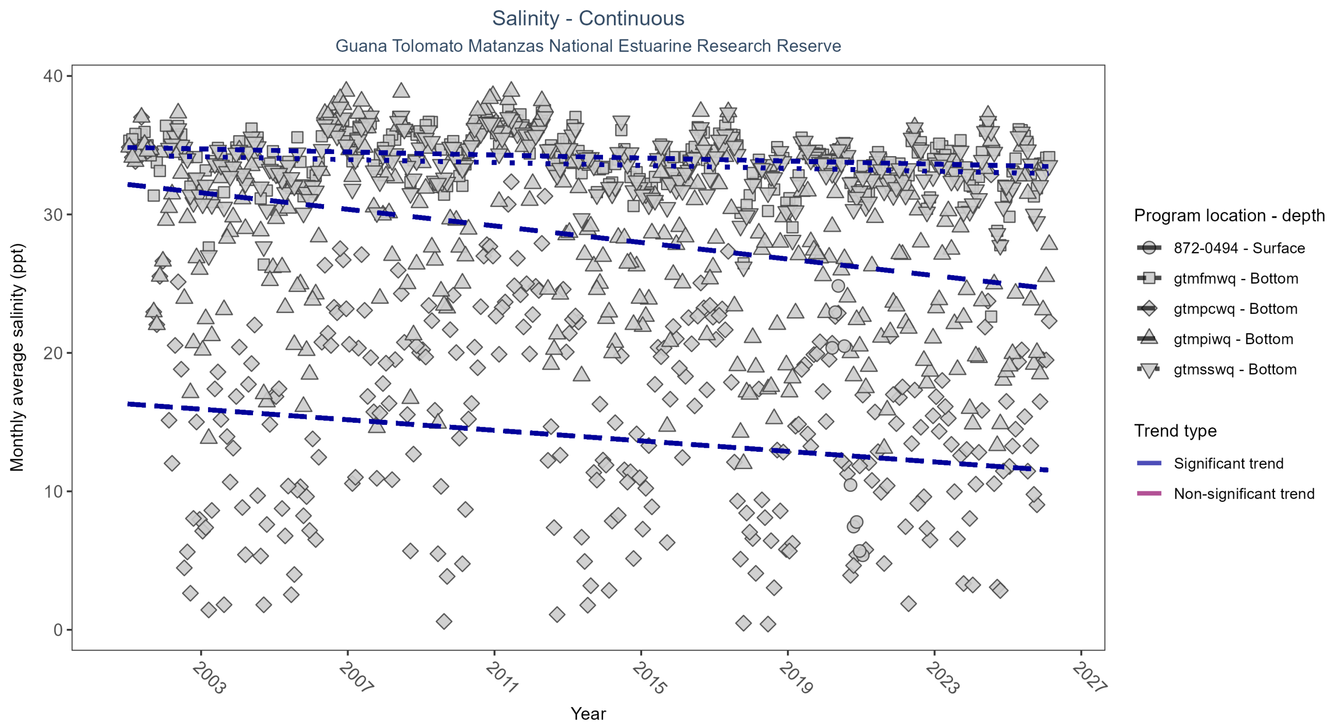

Figure 29: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 34: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	703518	26	2001 - 2026	27.70	-0.26	32.16	-0.3	0
872-0494	Insufficient data to calculate trend	34918	2	2020 - 2021	8.99	-	-	-	-
gtmfmwq	Significantly decreasing trend	705987	26	2001 - 2026	34.40	-0.16	34.84	-0.05	0
gtmsswq	Significantly decreasing trend	683666	25	2002 - 2026	33.90	-0.12	34.22	-0.05	0
gtmpcwq	Significantly decreasing trend	745868	26	2001 - 2026	16.60	-0.12	16.31	-0.19	0

At four program locations, monthly average salinity decreased between 0.05 and 0.3 ppt per year. There was insufficient data to fit a model for one location.

Specific Conductivity - Continuous

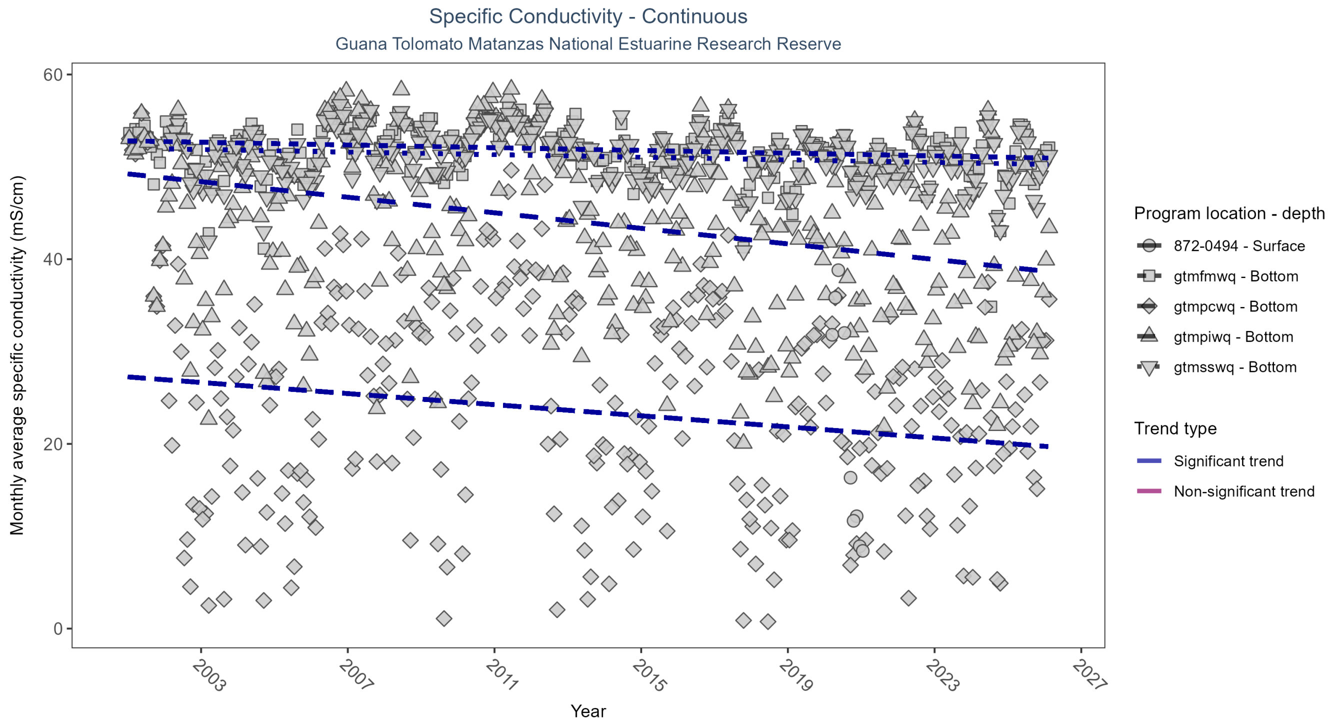

Figure 30: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 35: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	703518	26	2001 - 2026	43.06	-0.27	49.24	-0.42	0
872-0494	Insufficient data to calculate trend	34894	2	2020 - 2021	14.10	-	-	-	-
gtmfmwq	Significantly decreasing trend	705986	26	2001 - 2026	52.22	-0.15	52.81	-0.07	0
gtmsswq	Significantly decreasing trend	683665	25	2002 - 2026	51.54	-0.12	51.95	-0.07	0
gtmpcwq	Significantly decreasing trend	746635	26	2001 - 2026	27.04	-0.13	27.25	-0.3	0

At four program locations, monthly average specific conductivity decreased between 0.07 and 0.42 mS/cm per year. There was insufficient data to fit a model for one location.

Coastal Wetlands

The data file used is: **All__CW__Parameters-2026-Jun-04.txt**

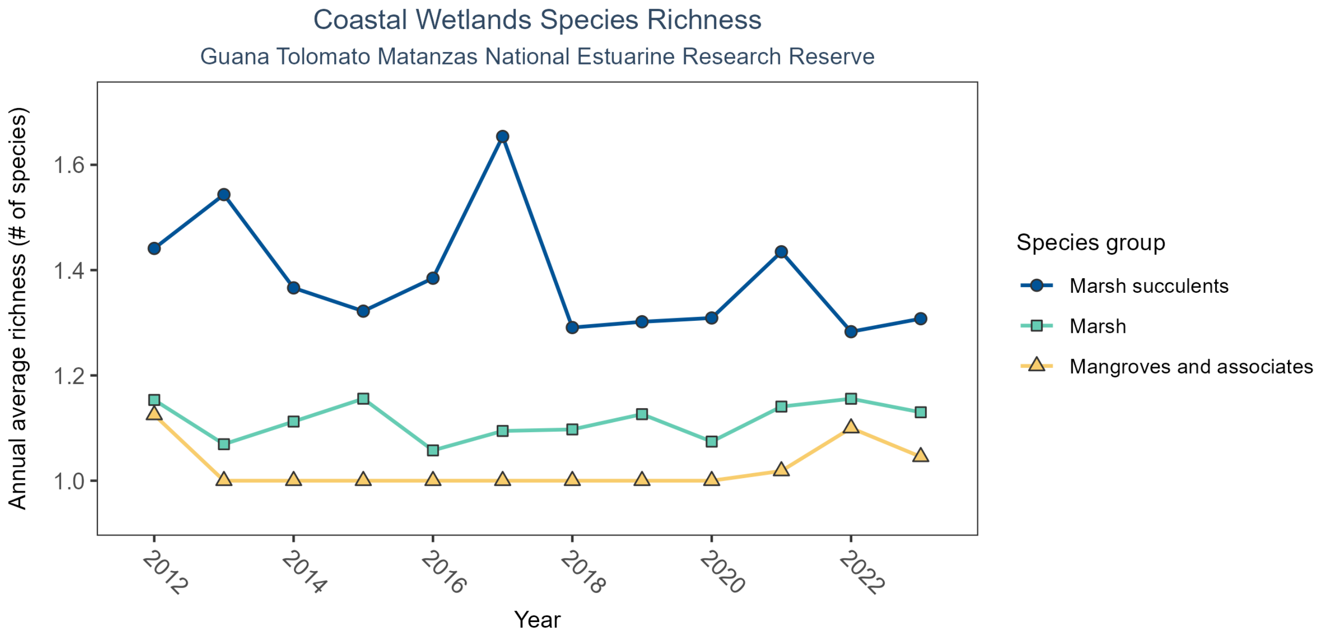

Figure 31: Line graph of annual average coastal wetlands species richness over time for mangroves and associates (triangles), marsh (squares), and marsh succulents (circles). If the time series by species group included more than one year of observations, a line connects data points for visualization.

Table 36: Coastal Wetlands Species Richness

Species Group	*No. of Samples*	*No. Years with Data*	*Period of Record*	*Median No. of Taxa*	*Mean No. of Taxa*
Mangroves and associates	387	12	2012 - 2023	1	1.02
Marsh	2568	12	2012 - 2023	1	1.12
Marsh succulents	868	12	2012 - 2023	1	1.38

Between 2012 and 2023, the median annual number of species for mangroves and associates was 1 based on 387 observations. Between 2012 and 2023, the median annual number of species for marsh was 1 based on 2,568 observations. Between 2012 and 2023, the median annual number of species for marsh succulents was 1 based on 868 observations.

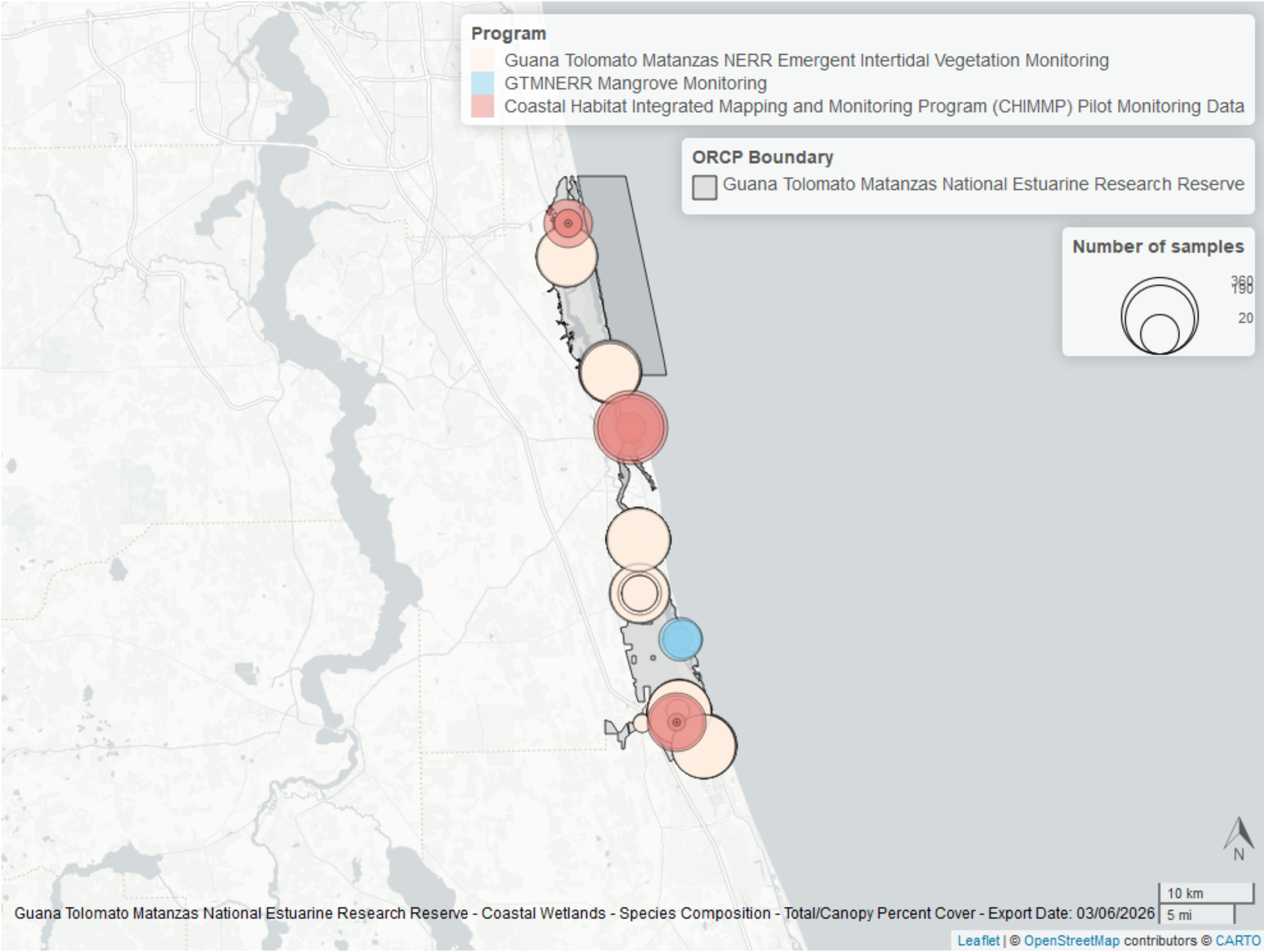

Figure 32: Map showing location of coastal wetlands sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

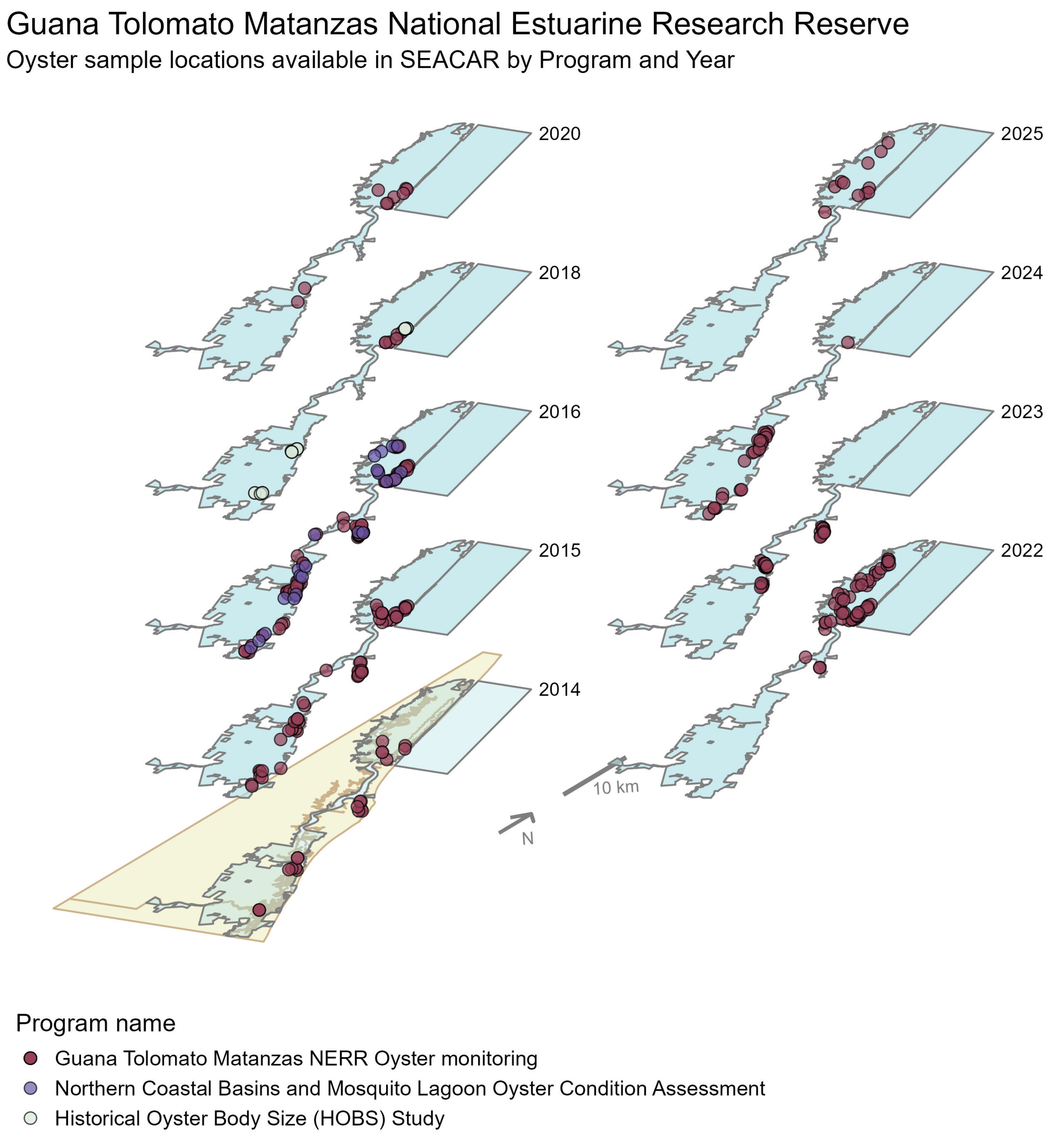

Figure 33: Maps showing the temporal scope of oyster sampling sites within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve* by program name.

Click here to view spatio-temporal plots on GitHub.

Density

For natural reefs, density decreased by an average of 137.22 oysters per square meter per year.

Natural

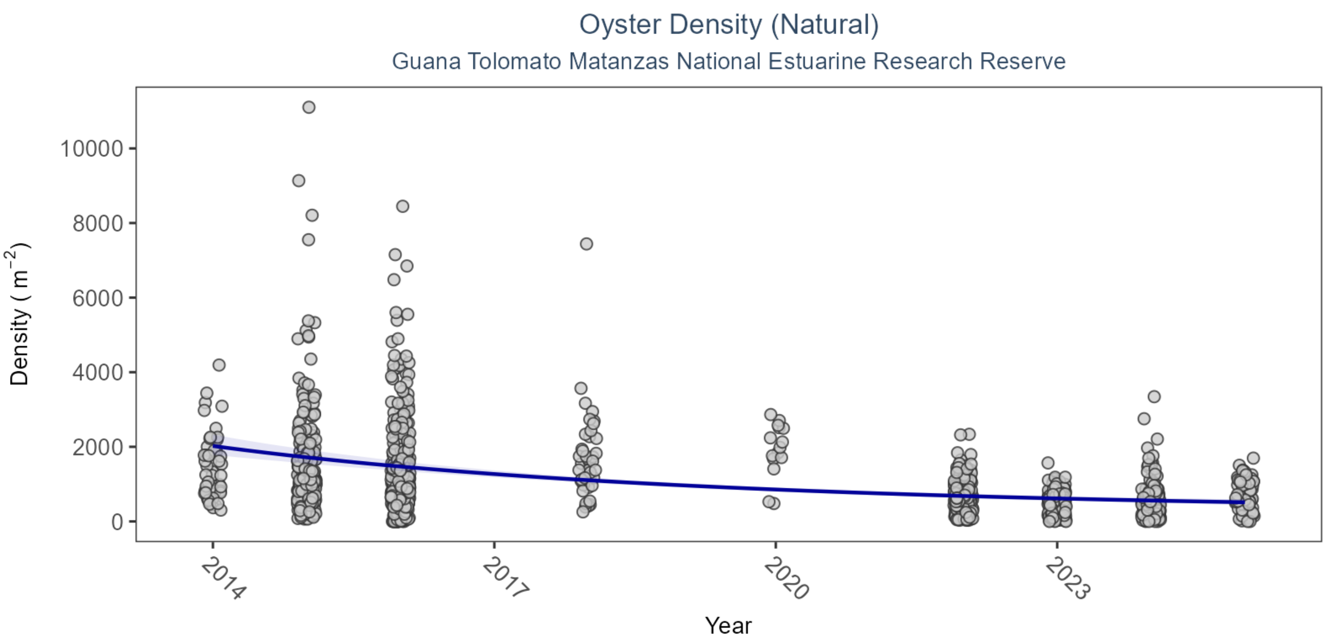

Figure 34: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 37: Model results for Oyster Density - Natural

Habitat Type	*Shell Type*	*Statistical Trend*	*Period of Record*	*Estimate*	*Standard Error*	*Credible Interval*
Natural	Live Oysters	Decreasing trend	2014 - 2025	-137.22	13.32	-166.31 to -113.69

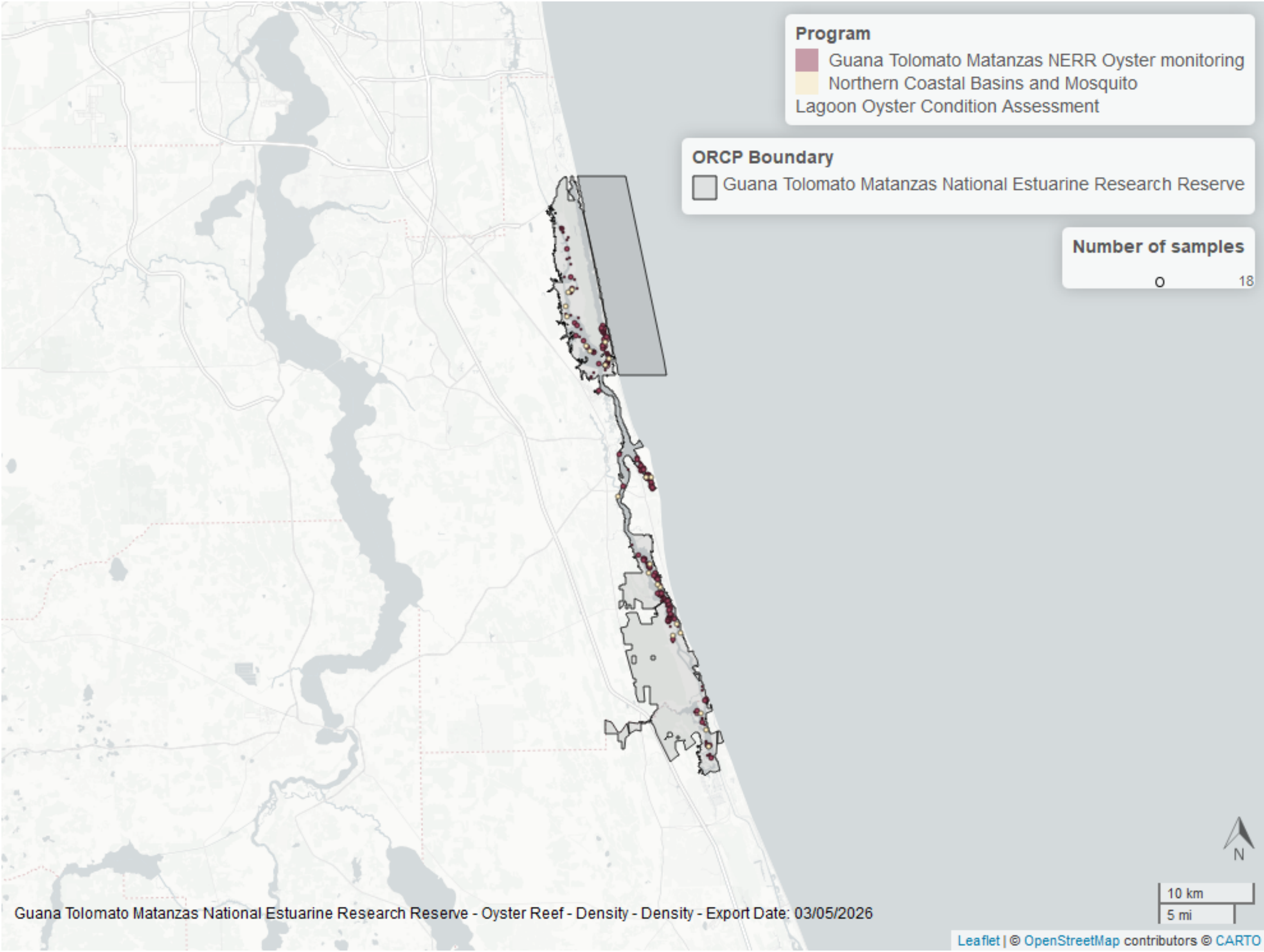

Figure 35: Map showing location of oyster density sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Percent Live

For natural reefs, percent live cover decreased by an average of 0.48% per year.

Natural

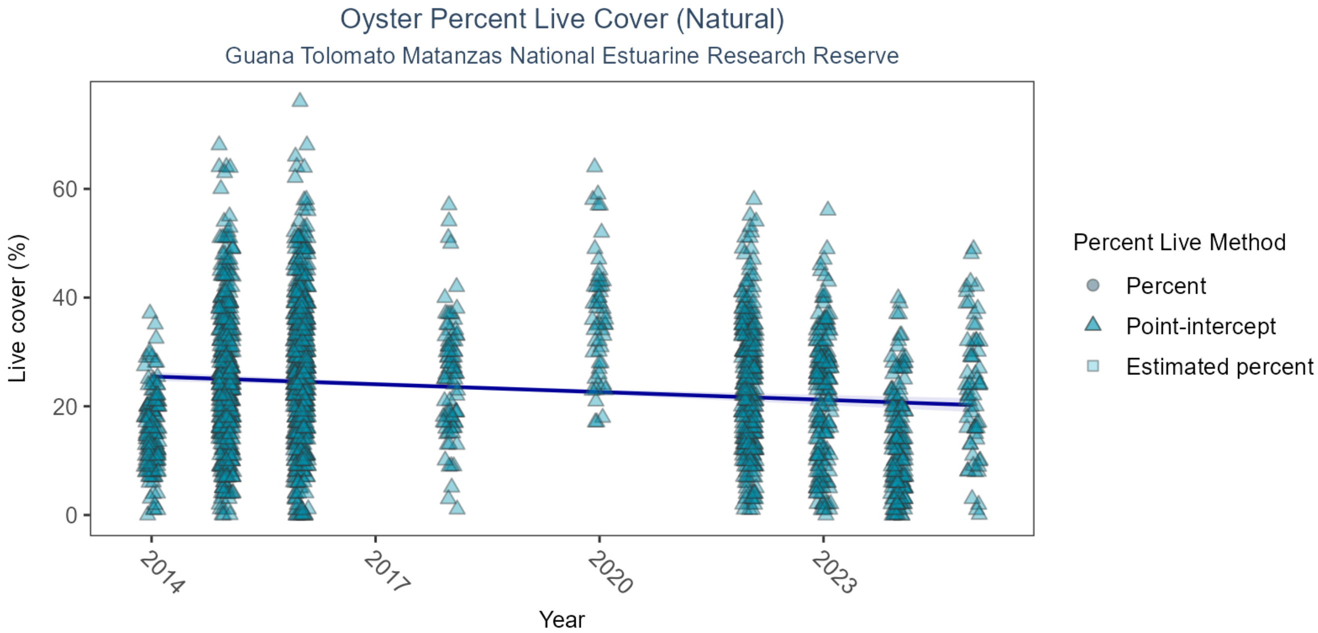

Figure 36: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 38: Model results for Oyster Percent Live - Natural

Habitat Type	*Shell Type*	*Statistical Trend*	*Period of Record*	*Estimate*	*Standard Error*	*Credible Interval*
Natural	Live Oysters	Decreasing trend	2014 - 2025	-0.48	0.09	-0.65 to -0.3

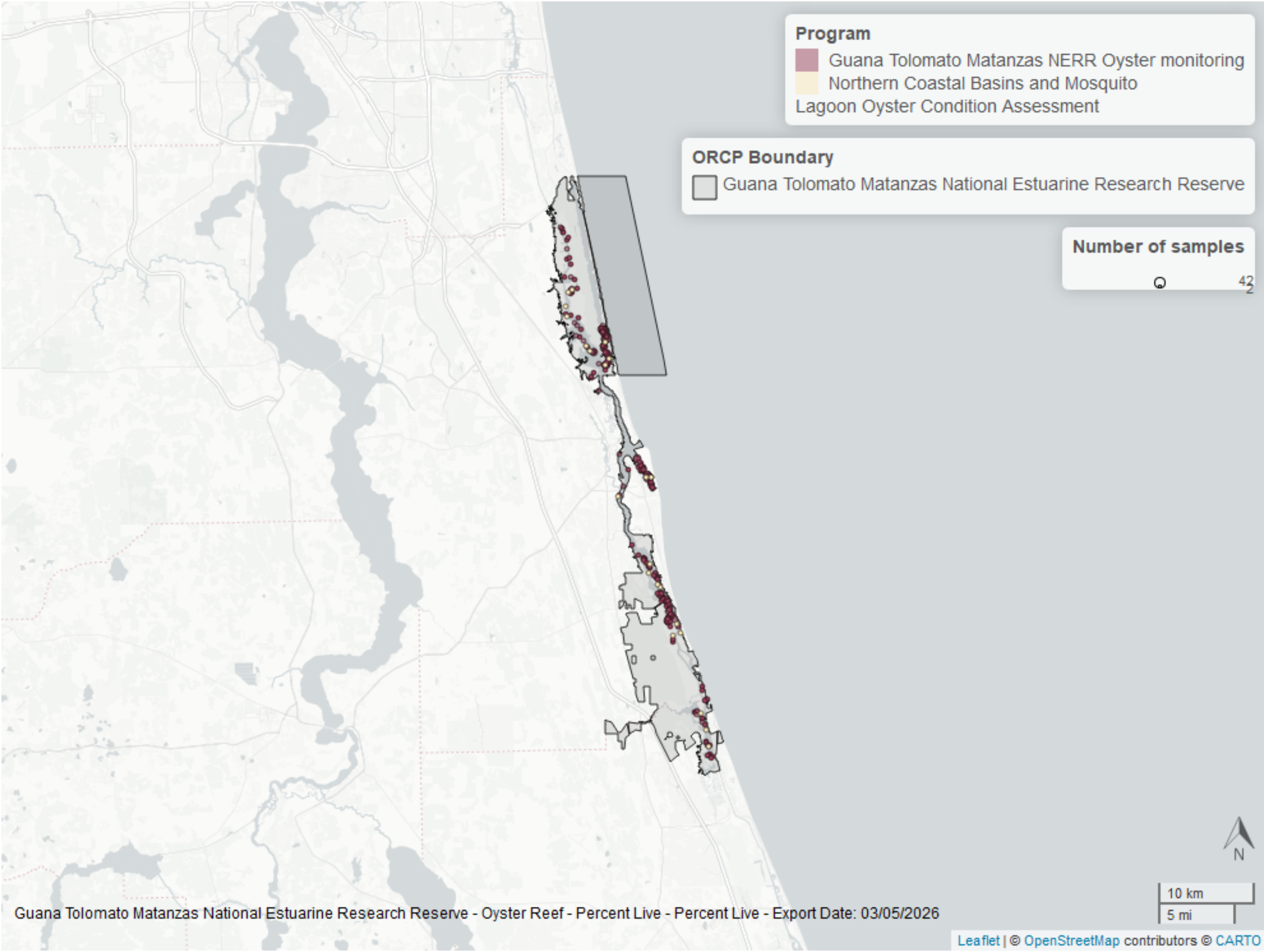

Figure 37: Map showing location of oyster percent live sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Shell Height

For natural reefs, annual average live oyster shell height in the 25-75mm size class increased by 0.26mm per year, and there was no detectable trend for live oysters in the >=75mm size class. Models are not run on dead oyster shell measurements.

Natural

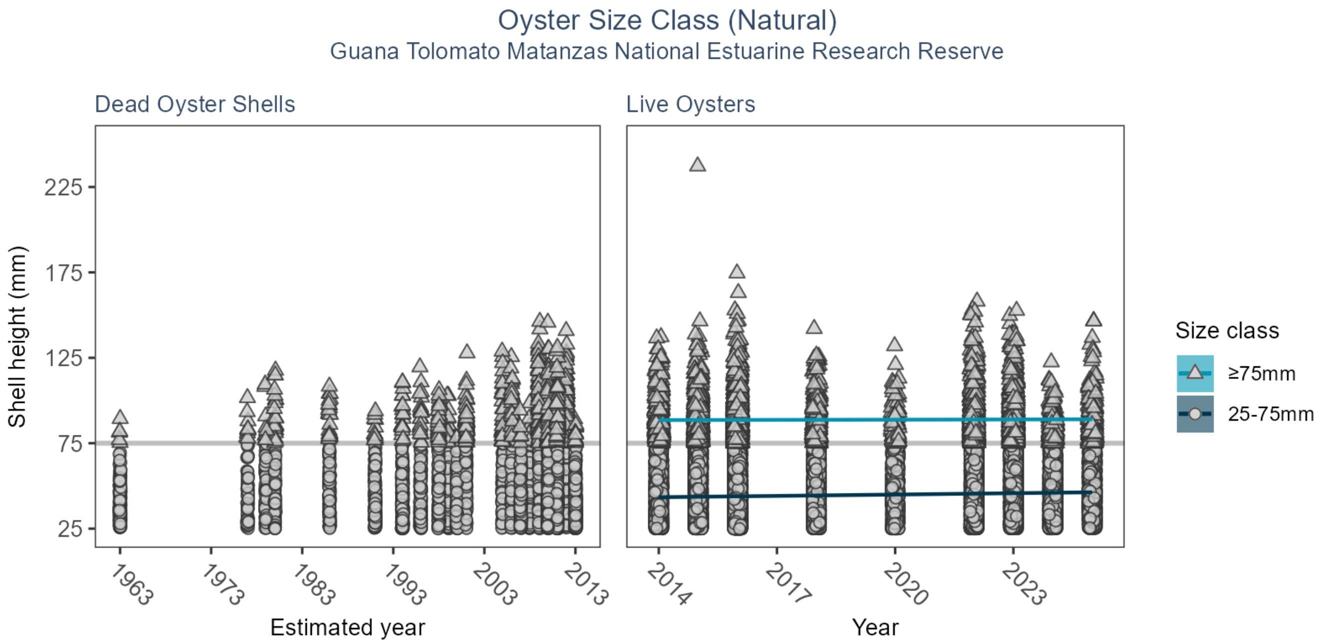

Table 39: Model results for Oyster Shell Height - Natural

Habitat Type	*Shell Type*	*SizeClass*	*Statistical Trend*	*Period of Record*	*No. of Samples*	*Estimate*	*Standard Error*	*Credible Interval*
Natural	Dead Oyster Shells	>75mm	Model not run on dead oyster shell	1749 - 2013	881	-	-	-
Natural	Dead Oyster Shells	25-75mm	Model not run on dead oyster shell	1749 - 2013	4639	-	-	-
Natural	Live Oysters	>75mm	No detectable trend	2014 - 2025	6155	0.04	0.07	-0.1 to 0.17
Natural	Live Oysters	25-75mm	Increasing trend	2014 - 2025	47020	0.26	0.03	0.2 to 0.32

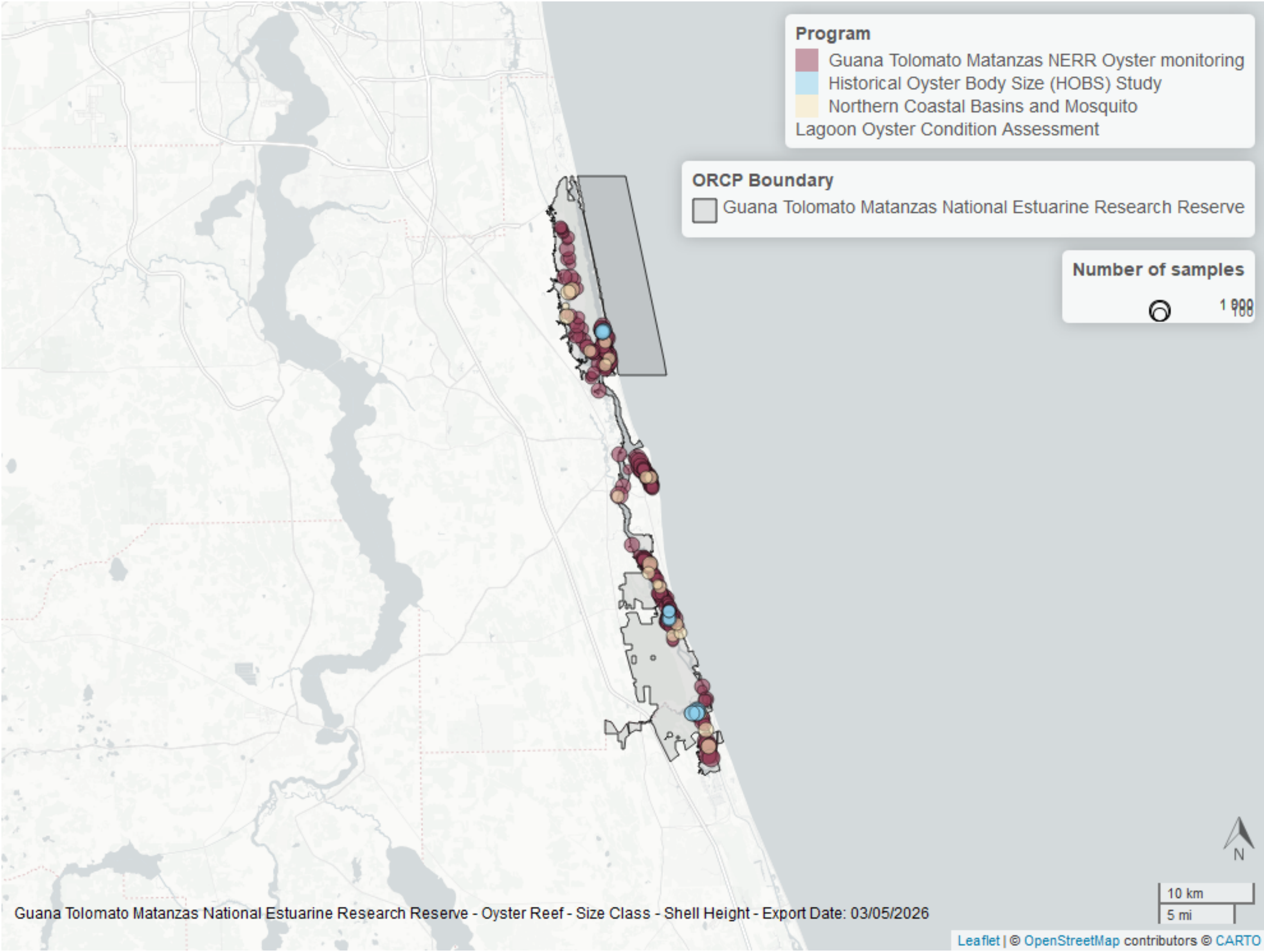

Figure 38: Map showing location of oyster shell height sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Acer rubrum	Hypericum sp.	Rhynchospora colorata
Acrostichum danaeifolium	Ilex vomitoria	Rhynchospora corniculata
Amorpha fruticosa	Ipomoea sagittata	Rhynchospora sp.
Ampelaster carolinianus	Iresine rhizomatosa	Rubus sp.
Ampelopsis arborea	Iva frutescens	Sabal palmetto
Andropogon sp.	Juncus roemerianus[1]	Sabatia calycina
Apios americana	Juniperus virginiana	Sabatia stellaris
Aristida sp.	Kosteletzkya pentacarpos	Sagittaria lancifolia
Avicennia germinans[1]	Laguncularia racemosa[1]	Sagittaria latifolia
Baccharis halimifolia	Limonium carolinianum[1]	Sagittaria subulata
Baccharis sp.	Lobelia cardinalis	Salicornia ambigua[1]
Bacopa monnieri	Ludwigia alata	Salicornia virginica[1]
Bare substrate	Ludwigia repens	Samolus ebracteatus
Batis maritima[1]	Ludwigia sp.	Samolus sp.
Blutaparon vermiculare[1]	Lygodium microphyllum	Samolus valerandi
Boehmeria cylindrica	Lythrum alatum	Serenoa repens
Borrichia frutescens	Lythrum lineare	Sesuvium portulacastrum[1]
Centella asiatica	Mikania scandens	Smilax bona-nox
Cicuta maculata	Myrica cerifera	Solidago sempervirens
Cladium mariscus	Nekemias arborea	Spartina alterniflora[1]
Coleataenia anceps	Osmunda regalis	Spartina bakeri[1]
Cyperus sp.	Osmundastrum cinnamomeum	Spartina patens[1]
Dichanthelium commutatum	Parthenocissus quinquefolia	Sporobolus virginicus[1]
Dichanthelium sp.	Pattalias palustre	Stenotaphrum secundatum
Dichondra carolinensis	Peltandra virginica	Symphyotrichum elliottii
Dichondra sp.	Persea palustris	Symphyotrichum tenuifolium
Distichlis spicata[1]	Persicaria hydropiperoides	Thelypteris kunthii
Eleocharis baldwinii	Persicaria punctata	Thelypteris palustris
Eleocharis geniculata	Persicaria sp.	Tillandsia usneoides
Eleocharis sp.	Pinus palustris	Toxicodendron radicans
Erechtites hieraciifolius	Pluchea odorata	Ulmus americana
Fabaceae	Pluchea sp.	Vigna luteola
Fimbristylis sp.	Pneumatophore	Vitis sp.
Fimbristylis spadicea	Polygonum sp.	Woody debris
Houstonia procumbens	Quercus virginiana	Acer rubrum
Hydrocotyle umbellata	Rhizophora mangle[1]	Acrostichum danaeifolium

1 - Coastal Wetlands

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